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Research Project Report on the Topic:

Diagnosis of Pathologies of the Cardiorespiratory System and Main Arteries Based on the Analysis of Sound Data

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Annotation

This work presents a reproducible comparative study of machine-learning methods for the acoustic screening of cardiorespiratory pathologies from auscultation sound. Two modalities are examined under one unified, leakage-safe pipeline: heart sounds (phonocardiograms) from PhysioNet/CinC 2016 and respiratory sounds from ICBHI 2017. A third modality, arterial bruits, is treated analytically, as no public dataset suitable for supervised learning exists. The central methodological contribution is a single shared evaluation protocol for both modalities: strictly grouped partitioning (patient-level for lung, recording-level for heart) preventing window-level leakage, and a shared balanced metric, the mean accuracy $\mathrm{MAcc} = (\mathrm{Se} + \mathrm{Sp})/2$, that makes the tasks directly comparable. Two families are contrasted: classical pipelines (MFCC with delta and spectral features fed to four standard classifiers) and deep learning on log-mel spectrograms (a compact CNN and an EfficientNet-B0, tuned by hyperparameter optimisation over 128 trials and reported as mean±std across three seeds). On heart sounds the best classical model (XGBoost, extended features) reaches MAcc = 0.903 (Se = 0.946, Sp = 0.859); on a recording-level split this may be optimistic. The tuned EfficientNet-B0 reaches 0.898 ± 0.008, comparable with it but not ahead. On lung sounds all methods perform within one narrow band; a fine-tuned Audio Spectrogram Transformer is the single best (ICBHI = 0.594 ± 0.023, three seeds), above EfficientNet-B0 (0.555 ± 0.016). A seed-robust cross-modal analysis finds that classical method rankings do not transfer between modalities, whereas deep encoders transfer benignly in both directions; a transfer asymmetry seen at a single seed did not survive multi-seed averaging. On the CirCor DigiScope 2022 dataset, with a patient-level split, deep learning overtakes the classical models on murmur detection (0.810 vs. 0.688), reversing the CinC ordering. A limitations section bounds the clinical claims.

Keywords

auscultation, phonocardiogram, respiratory sound, machine learning, deep learning, MFCC, convolutional neural network, grouped-split evaluation, data leakage, comparative study.

Аннотация

В работе представлено воспроизводимое сравнительное исследование методов машинного обучения для акустического скрининга патологий кардиореспираторной системы по звукам аускультации. В рамках единого, устойчивого к утечке данных конвейера рассматриваются две модальности: тоны сердца (фонокардиограммы) из PhysioNet/CinC 2016 и дыхательные шумы из ICBHI 2017. Третья модальность — артериальные шумы — рассмотрена аналитически: открытого размеченного набора для обучения с учителем не существует. Ключевой методический вклад — единый протокол оценки для обеих модальностей: строгое групповое разбиение, исключающее утечку данных (на уровне пациента для лёгких и записи для сердца), фиксированные начальные значения генераторов, закреплённые версии библиотек и единое семейство метрик ($(\mathrm{Se} + \mathrm{Sp})/2$), делающее задачи напрямую сопоставимыми. Сравниваются классические конвейеры (мел-кепстральные коэффициенты с дельта-признаками и спектральными статистиками, поданные в четыре стандартных классификатора) и глубокое обучение на лог-мел спектрограммах (компактная свёрточная сеть и трансферная модель EfficientNet-B0, настроенные оптимизацией гиперпараметров и усреднённые по трём прогонам с разной инициализацией). На тонах сердца лучший результат — MAcc = 0.903 (XGBoost, Se = 0.946, Sp = 0.859; разбиение по записям, возможно оптимистично); EfficientNet-B0 достигает 0.898 ± 0.008 — сопоставимо, но классическая модель остаётся численно лучшей. На дыхательных шумах все методы находятся в одном диапазоне (ICBHI ≈ 0.54–0.59); лучший результат — у дообученного Audio Spectrogram Transformer (0.594 ± 0.023), выше EfficientNet-B0 (0.555 ± 0.016). Межмодальный анализ по трём прогонам показывает: ранги классических методов между модальностями не переносятся, тогда как глубокие энкодеры переносятся доброкачественно в обе стороны (асимметрия одного прогона не пережила усреднение). На датасете CirCor DigiScope 2022 с разбиением по пациентам глубокое обучение обгоняет классику в детекции шумов (0.810 против 0.688), обращая порядок CinC.

Ключевые слова

аускультация, фонокардиограмма, дыхательный шум, машинное обучение, глубокое обучение, MFCC, свёрточная нейронная сеть, групповое разбиение, утечка данных, сравнительное исследование.

Introduction

Auscultation (listening to the internal sounds of the body with a stethoscope) remains one of the oldest and most accessible diagnostic techniques in medicine. Despite the spread of imaging, the acoustic signatures of the heart, lungs and large arteries still carry clinically decisive information: a murmur can reveal a valvular defect, a crackle or wheeze can localise pulmonary pathology, and a bruit can betray a narrowed carotid artery. Yet the skill of interpreting these sounds is unevenly distributed, hard to acquire and subjective. Automated analysis of auscultation recordings therefore promises a low-cost screening and triage aid, particularly where specialist expertise is scarce.

This project studies, empirically and comparatively, how well machine learning can support such a screening task across more than one auscultation modality at once. We deliberately frame the contribution as a screening and triage aid rather than a diagnostic instrument: the goal is to flag recordings that warrant expert review, not to replace the clinician.

Relevance

Three factors make the problem timely. First, large, openly licensed datasets of heart and lung sounds now exist, removing the historical barrier of data scarcity for these two modalities. Second, the methodological pitfalls of audio classification (chiefly data leakage from splitting recordings rather than patients) are now well documented, so a study that avoids them produces results that are genuinely comparable across methods. Third, most publicly available work treats each modality in isolation; a single pipeline evaluated under one shared protocol on heart and lung sounds is comparatively rare, and arterial sounds are almost entirely unaddressed for want of open data.

Practical significance

A reliable acoustic screening tool would let non-specialists capture a short recording and obtain an objective “normal / needs review” indication. In primary care, remote regions and tele-medicine this could shorten the path to a specialist for patients who need one while reducing unnecessary referrals. Because the same software backbone serves several modalities, the practical cost of deploying it across heart and lung screening is shared rather than duplicated.

Innovation

The novelty of this study is a *unified, leakage-safe, cross-modal comparison pipeline*. A single codebase, governed by per-modality configuration files, applies the same preprocessing, the same leakage-safe grouped partitioning (patient-level for lung, recording-level for heart) with an explicit zero-leakage assertion, the same metric family and the same set of classical and deep models to both heart and lung sounds. This lets us ask a question that single-modality studies cannot: do the method families that win on one auscultation modality also win on another? We further contribute an honest analytical treatment of arterial bruits, documenting precisely why supervised learning is not yet feasible there and how the same pipeline would extend if data became available. The emphasis throughout is on reliability and reproducibility (honest baselines under a leakage-free protocol) rather than on chasing state-of-the-art accuracy through aggressive tuning.

Concretely, the work makes five contributions:

1. a single leakage-safe pipeline that applies the same preprocessing, feature extraction and modelling, with a grouped, leakage-safe partitioning and an explicit zero-leakage assertion, to two auscultation modalities, enabling a like-for-like comparison that isolated single-modality studies cannot offer;
2. a head-to-head comparison of the classical family (MFCC features with logistic regression, support-vector machines, random forests and gradient boosting) against the deep family (a compact convolutional network and a hyperparameter-tuned, three-seed EfficientNet-B0) under one class-imbalance-robust metric per task;
3. a seed-robust cross-modal analysis showing that classical method *rankings* do not transfer between modalities — the best model on heart is not the best on lung, a non-transfer that survives a nine-classifier panel, is confirmed by a paired McNemar test, and traces to a model-family-by-modality interaction. Deep encoders, by contrast, transfer benignly in both directions under fine-tuning, and a transfer asymmetry suggested by a single seed does not survive multi-seed evaluation — a result invisible to single-modality and single-seed studies alike;
4. an extension to a third heart dataset (CirCor DigiScope 2022) on a genuinely patient-level split, showing that the classical-versus-deep verdict is task-dependent: deep learning overtakes the classical models on the finer murmur-detection task, reversing their order on the coarse normal/abnormal screen; and
5. an analytical arterial sub-study that states plainly why supervised learning is not yet possible for carotid bruits and specifies what an open dataset would need to provide, complemented by a synthetic proof-of-concept that runs the pipeline end-to-end on arterial-band audio.

Research objectives

The work pursues the following objectives:

1. to assemble a reproducible, leakage-safe experimental pipeline shared by the heart and lung modalities, with fixed seeds and pinned dependencies;
2. to extract classical acoustic features (MFCC with deltas and spectral statistics) and to train and evaluate logistic regression, support-vector machines, random forests and gradient boosting on both modalities;
3. to train deep-learning models (a compact convolutional network and an EfficientNet-B0 transfer model) on log-mel spectrograms of the same data;
4. to evaluate every model with the official, class-imbalance-robust metric of each task and with confusion matrices, aggregating to the clinically meaningful recording level for heart sounds;

5. to analyse how method rankings transfer across the two modalities, and to situate arterial-sound diagnosis analytically; and
6. to deliver an original, fully cited research report conforming to the department's structural and formatting requirements.

The remainder of the report is organised as follows. Chapter 1 reviews the physiological basis, the datasets and the prior classical and deep-learning approaches, and identifies the research gap. Chapter 2 specifies the methodology in full. Chapter 3 reports the experimental results for heart and lung sounds. Chapter 4 presents the cross-modal analysis and the arterial sub-study. The Conclusion summarises the findings against each objective and states the limitations and directions for future work.

Project group and contribution

This work is a *group (team) research project* in the department's sense, though the project group in this instance consists of a single member, the author. Accordingly, the contribution is undivided: all stages of the work were performed by the author, Tsember Andrei Alekseevich (group БПАД244): problem formulation and the analytical literature review; acquisition and exploratory analysis of the PhysioNet/CinC 2016 (heart) and ICBHI 2017 (lung) datasets; preprocessing, segmentation and the construction of leakage-safe grouped splits; classical feature engineering and the logistic-regression, support-vector-machine, random-forest and gradient-boosting experiments; the convolutional-network and EfficientNet-B0 deep-learning experiments, including the hyperparameter search and multi-seed evaluation, together with the integration and fine-tuning of an Audio Spectrogram Transformer; the cross-modal transfer analysis and the arterial analytical sub-study; and the preparation of this report under the department's structural and formatting requirements.

1 Analytical review of existing approaches

This chapter positions the present study within the literature on auscultation- sound classification. It proceeds from the physiological basis of the signals, through the public datasets that enable reproducible benchmarking, through the two dominant method families (classical feature engineering and deep learning), to the cross-modality question that motivates our contribution. The chapter closes with an explicit statement of the research gap and the novelty claim.

1.1 Physiological basis of auscultation sounds

Auscultation has been a cornerstone of cardiovascular and pulmonary diagnosis since Laënnec introduced the stethoscope in the early nineteenth century [1]. The sounds produced by the body carry information about the mechanical state of organs and vessels in a form that is non-invasive, low-cost and immediately available at the point of care. Understanding the acoustic origins of each modality is prerequisite to designing appropriate signal-processing chains.

1.1.1 Heart sounds

Normal cardiac auscultation yields two dominant events per cycle. The first heart sound (S1) coincides with closure of the mitral and tricuspid valves at the onset of systole; the second heart sound (S2) arises from aortic and pulmonic valve closure at the onset of diastole. Both events generate transient, broadband pressure waves in the 20–400 Hz range [2]. Pathological states alter this pattern in characteristic ways: valve stenosis or regurgitation introduces turbulent flow and produces sustained murmurs occupying the systolic or diastolic interval; cardiomyopathies may produce third (S3) or fourth (S4) sounds; congenital defects can produce fixed splitting of S2. Because the clinically diagnostic content lies largely in the 20–400 Hz band, a bandpass filter with those cutoffs suffices for most classification tasks while effectively removing low-frequency movement artefact and ambient noise above the relevant range.

1.1.2 Respiratory sounds

Breath sounds originate in turbulent airflow through the tracheobronchial tree. In health that turbulence is heard as the soft, low-pitched vesicular murmur of inspiration; disease superimposes additional, so-called adventitious, sounds upon it. Two adventitious types dominate the clinical picture. Crackles, which the older literature calls rales, are brief, intermittent, popping noises usually explained by alveoli or small airways snapping back open after collapse, and they accompany pneumonia, pulmonary fibrosis and cardiac failure. Wheezes are their opposite in character: sustained, tonal and high in pitch, generated when the walls of a narrowed airway flutter, and typical of obstructive disease such as asthma and chronic obstructive pulmonary disease (COPD) [1]. The ICBHI annotation scheme mirrors exactly this clinical taxonomy, sorting each cycle into one of four categories: normal, crackle, wheeze, or “both” (crackle and wheeze together) [3]. Diagnostically relevant energy spans roughly 200–1800 Hz, with crackles biased toward the upper part of that band and wheezes toward 200–600 Hz [1].

1.1.3 Arterial bruits

A carotid bruit is a systolic murmur-like sound produced when blood accelerates through a stenosed region of the carotid artery, generating turbulent flow audible with a stethoscope placed over the neck. The acoustic signature overlaps with cardiac murmurs in bandwidth but is shorter in duration and strongly modulated by the stenosis severity. The clinical value of bruit detection as a screening marker for carotid artery disease is well established; pilot systems have demonstrated that even smartphone microphones can capture diagnostically relevant features [4]. However, no openly licensed, cycle-labelled dataset of carotid bruit recordings has been released to date, which precludes a supervised-learning study for this modality [5]. We therefore treat arterial sounds analytically in Chapter 4.

1.2 Public datasets for auscultation research

The availability of large, openly licensed auscultation datasets is a prerequisite for reproducible benchmarking. Two collections have emerged as the community standards for heart and lung sounds respectively, and they are the primary data sources for this study.

1.2.1 PhysioNet/CinC 2016: heart sounds

The 2016 PhysioNet/Computing-in-Cardiology Challenge distributed a public training pool assembled from six separately collected cohorts, conventionally labelled A–F; pooled, these supply on the order of 3240 phonocardiograms originating from 764 distinct individuals [2,6]. The present study draws on the five cohorts A–E (3126 recordings); the small F cohort is left out. Clip durations vary widely (the shortest last a few seconds, the longest around two minutes) because the audio was captured with consumer-grade electronic stethoscopes in working clinics and homes rather than under laboratory control, a choice that deliberately exposes a model to the ambient noise and acquisition variability of real deployment. Each clip carries one of two verdicts, normal or abnormal, alongside a small residual “unsure” tier that the organisers advised discarding or folding into the negative class. The labels are far from balanced (normals account for roughly seven recordings in ten and abnormals for fewer than one in five) which is why the challenge scores submissions not on raw accuracy but on the mean of the true-positive and true-negative rates, the mean accuracy (MAcc), a figure of merit that refuses to let the dominant normal class mask poor detection of the abnormal one. The strongest challenge entries reached MAcc in the region of 0.86 [7], which stands as the public reference point for the task.

A methodological subtlety follows from the release design: only the training set was published with labels, while the test set was kept private. Anyone using the corpus must therefore carve their own train/test partition out of the training pool, and to avoid leakage that partition should ideally separate subjects rather than recordings [8]. In practice the public release ships no complete recording-to-subject mapping, so studies commonly group by recording instead, a limitation that the present work states explicitly in Chapter 2. This distinction separates rigorous studies from a body of work that reports inflated figures because several recordings from one patient land on both sides of a random split.

The CirCor DigiScope 2022 database [9] extends the phonocardiogram landscape with 5272 recordings from 1568 subjects, adding graded murmur metadata (timing, shape, pitch, grade) that enables finer-grained classification. We note it here as a natural extension for future work but do not use it in this study because the scope is binary normal/abnormal heart discrimination, where CinC 2016 is the established benchmark.

1.2.2 ICBHI 2017: respiratory sounds

The ICBHI 2017 Respiratory Sound Database, released through the International Conference on Biomedical and Health Informatics [3], gathers 920 audio files from a cohort of 126 individuals. Expert annotation operates at the granularity of the individual breath cycle rather than the whole recording, yielding 6898 labelled cycles. Acquisition used four different devices, each digitising at its own rate (4 kHz at the low end and 44.1 kHz at the high end, with 10 kHz and 22.05 kHz in between), so a resampling stage is needed to reconcile their differing spectral characteristics. The cycles are spread unevenly across the four labels: just over half are normal, crackle-only cycles make up a little more than a quarter, isolated wheezes about an eighth, and the combined crackle-and-wheeze category is rarest at well under a tenth. A further contrast with CinC 2016 is that the organisers fix the partition themselves: close to three-fifths of the subjects form the training pool and the remainder the test pool, with no subject straddling the boundary. Honouring this prescribed split is essential for comparability with prior work; substituting a random, recording-level split can lift the reported ICBHI score by several points [8].

The ICBHI score (the same balance of sensitivity and specificity, but with sensitivity aggregated over abnormal cycles and specificity measured on normal cycles) mirrors the CinC MAcc in structure, making the two tasks directly comparable under a single metric family. This alignment is a design choice exploited in the present study.

1.3 Classical feature-based approaches

Prior to the deep-learning era, and still competitive for modest dataset sizes, the dominant paradigm was to extract a fixed-dimensional acoustic feature vector from each audio segment and to train a standard classifier on those vectors.

1.3.1 Feature representations

By far the most common engineered representation in this field is the mel-frequency cepstral coefficient (MFCC), a device borrowed from automatic speech analysis [10] and transferred to cardiac and pulmonary audio because those signals, like speech, are band-limited and approximately stationary over short frames. The computation re-expresses each frame's energy across a bank of bands spaced to mimic perceived pitch, takes the logarithm of those band energies, and then applies an orthogonalising (discrete-cosine) transform that removes the correlation between neighbouring bands; only the first few dozen outputs are kept (here of order 13–40) and together they summarise the spectral envelope of a heart or lung event. Adding the first- and second-order frame-to-frame differences (Δ and $\Delta\Delta$) records how that envelope moves over time, which bears directly on the onset and decay of murmurs and on the sharp transients of crackles. Because clip lengths differ, each coefficient's per-frame series is finally collapsed to two summary statistics (its average and its spread across frames) producing a descriptor whose dimensionality no longer depends on recording duration; vectors built on exactly this construction carried a submission into the top tier of the CinC 2016 challenge [7].

We additionally compute a small set of global spectral-shape descriptors: the spectral centroid (the spectral centre of mass), the 85%-energy roll-off frequency, bandwidth, zero-crossing rate and root-mean-square (RMS) energy. These broadband descriptors are inexpensive to compute and characterise the distribution of energy that the cepstral coefficients capture only indirectly; whether they add discriminative value beyond the MFCC block is one of the questions our feature-set comparison (A versus B) is designed to answer empirically.

1.3.2 Classifiers

Among classifiers fitted to these descriptors, the support-vector machine with a Gaussian (radial-basis-function) kernel [11] is reported most often: the kernel lets the model trace curved decision surfaces in the original feature space without ever materialising the high-dimensional space those surfaces formally inhabit, which suits cepstral classes whose boundaries are not linear, and it stays numerically stable in the regime typical here, with hundreds of features and tens of thousands of segments. Two tree ensembles serve as the principal alternatives. A random forest [12] grows many trees on perturbed views of the data and pools their votes, damping the variance of any single tree; gradient boosting, in its XGBoost form [13], instead grows trees in sequence so that each one corrects its predecessors' residual errors, and it frequently matches or beats the SVM at lower inference cost while exposing per-feature importances that lend a degree of interpretability. Logistic regression is kept as a deliberately transparent linear yardstick: it cannot represent feature interactions, but the cepstral space is often well-conditioned enough for a linear boundary to compete, which makes it a principled lower bound against which the non-linear models are judged.

A concern shared by all of these classifiers is class imbalance. One widely used remedy is to synthesise minority examples with SMOTE [14], but only ever inside the training fold, because generating synthetic points before the train/test split leaks test-neighbour information into training [8]. The present study sidesteps synthesis altogether and instead reweights the per-class loss within each training fold (Chapter 2), which addresses the same majority-class bias without manufacturing data; the in-fold-only discipline is a recurring correctness requirement that a substantial fraction of published studies violate by resampling globally.

1.4 Deep-learning approaches

Deep learning on time–frequency images is now the prevailing approach to audio classification. For auscultation the customary input is a log-mel spectrogram, in effect a single-channel image in which one axis advances through time, the other climbs a perceptually scaled frequency ladder, and intensity encodes log-power. Such an image is detailed enough to expose both the spectral fingerprint of a murmur and the temporal rhythm of adventitious lung sounds, and (being image-shaped) it lets convolutional weights pre-trained on natural photographs be repurposed for audio.

1.4.1 Convolutional networks

Compact convolutional networks (CNNs) with three to five convolutional layers are a popular first deep-learning baseline because they train in minutes on a GPU and in tens of minutes on a CPU. A representative ICBHI study by Demir et al. converted respiratory recordings to short-time-Fourier spectrograms, used a pre-trained convolutional network as a fixed feature extractor, and classified the resulting embeddings with an SVM, a hybrid that outperformed purely hand-crafted baselines on the respiratory task [15]. This pattern, in which a convolutional backbone supplies the representation and a lightweight classifier makes the decision, recurs throughout the auscultation literature and directly motivates the transfer-learning approach examined next.

1.4.2 Transfer learning

A frequent transfer route adopts EfficientNet-B0 as the feature extractor [16]: its parameter budget (about four million) is small enough to fine-tune on a few thousand recordings yet large enough to inherit useful visual priors from ImageNet pretraining. Because the network expects three input planes, the single mel channel is tiled threefold (or routed through a thin projection) to match. A common fine-tuning schedule keeps the pretrained weights frozen for a brief warm-up and then releases them for end-to-end training, a recipe that pays off most when the audio corpus is small, on the order of a few thousand recordings, and the backbone's image priors supply helpful inductive bias.

1.4.3 Audio transformers and recent advances

The Audio Spectrogram Transformer (AST) [17] removes convolution from the pipeline altogether: the mel image is cut into a grid of partly overlapping tiles, each tile becomes a token, and stacked self-attention layers learn how every tile relates to every other, giving the model a global view of spectro-temporal structure that a shallow CNN's limited receptive field cannot reach. Built on such a backbone, patch-mix contrastive training reported a new best result on the ICBHI benchmark [18]. These models are GPU-hungry, however, and frequently augment data before the train/test split, which can inflate published figures; indeed, a careful reading shows that many high-scoring respiratory papers quietly use random or recording-level partitions rather than the official patient-independent one, rendering their headline numbers incomparable with protocol-faithful studies [8].

1.5 The cross-modality gap and research novelty

The overwhelming majority of auscultation machine-learning research is unimodal: it addresses either heart sounds or lung sounds, but not both simultaneously. Yet a clinician uses the same instrument and a closely related interpretive framework across modalities; a computational tool that generalises across modalities is both more practical and more theoretically interesting. Unified models that operate on heart and lung sounds within a single pipeline are only beginning to be explored [19]. A recent generalist body-sound foundation model [20] represents an ambitious attempt at unification across auscultation modalities, but it relies on large-scale data not assembled for the community benchmark tasks.

Arterial bruits occupy a still-harder position: despite growing clinical interest in non-invasive carotid stenosis screening [4], no openly licensed dataset of labelled bruit recordings has been released, and the systems that have been evaluated rely entirely on proprietary collections [5]. This data gap prevents empirical supervised learning for arterial sounds and is likely to persist until a major institution commits to an open release comparable to PhysioNet or ICBHI.

Methodological reviews of machine learning for biomedical audio [8] identify three recurring pitfalls that prevent results from being trustworthy or reproducible: (i) recording-level rather than patient-level splitting; (ii) global data augmentation applied before the split, leaking augmented test-set neighbours into training; and (iii) inconsistent metric choices (raw accuracy on an imbalanced dataset vs. the balanced mean of sensitivity and specificity). All three pitfalls are present in a substantial fraction of the published auscultation literature.

1.5.1 Research gap and novelty claim

The research gap addressed by this study is therefore twofold. First, a single, leakage-safe pipeline evaluated under one shared protocol on both heart and lung sounds (same preprocessing, feature extraction, model set and balanced metric) is comparatively rare. Second, the question of whether method rankings transfer across auscultation modalities (does the best heart classifier also win on lung sounds?) is largely unstudied, even though the answer has practical implications for how much can be shared in a multi-modality system.

This study's novelty is a *unified, reproducible, cross-modal comparison* under a zero-leakage grouped-split protocol: the same codebase, preprocessing chain, model set and evaluation metric are applied to both CinC 2016 heart data and ICBHI 2017 lung data, enabling a direct, fair comparison of method rankings across modalities. The contribution is intentionally framed as honest leakage-free baselines rather than state-of-the-art scores, because the latter without the former is unreliable. An analytical treatment of arterial bruits documents precisely why supervised learning is not yet feasible and what would be required to extend the pipeline to a third modality.

1.5.2 Chapter summary

The literature offers mature single-modality methods (MFCC-based classical classifiers and CNN/transfer deep models) but few studies enforce leakage-safe grouped splits or compare methods across modalities under a single protocol. The cross-modal ranking question is largely open. This motivates the unified pipeline specified in Chapter 2, and the cross-modal analysis of Chapter 4.

2 Methodology

This chapter specifies the experimental pipeline in enough detail to reproduce every reported number. The design principle is *one shared, configuration-driven codebase* applied to both modalities through a common set of stages, with all randomness fixed and all dependency versions pinned. The pipeline is deliberately uniform across modalities. Only two protocol parameters differ between heart and lung — the grouping granularity and the test fraction — and both are forced by the datasets rather than chosen freely (Section 2.1). The pipeline is organised as independent stages (ingest, preprocess, segment, feature extraction, grouped split, training and evaluation), each writing its output to disk so any single stage is re-runnable in isolation.

2.1 Datasets and leakage-safe splits

Heart sounds are taken from the PhysioNet/CinC 2016 training databases (subsets A through E); respiratory sounds from the ICBHI 2017 database. The two raw collections are indexed into a single recording-level manifest carrying, for each recording, a grouping identifier (the patient where known, the recording otherwise), a class label, the modality and the file path; a separate cycle-level table records the annotated respiratory cycles of ICBHI with their start and end times and one of four labels.

The single most important correctness requirement is *grouped* (not random) partitioning, so that no recording (and, where subject identity is recoverable, no patient) appears on both sides of the split; this prevents the data leakage that inflates many published results. For heart sounds the split is a seeded `GroupShuffleSplit` (test fraction 0.20, random state 42) computed *within* databases A–E. The public CinC 2016 training release does not publish a complete recording-to-subject mapping, so the grouping key is the recording identifier: this guarantees that no 3-second window of a recording can fall on both sides of the split (the dominant leakage risk for windowed audio) but the resulting partition is recording-level rather than strictly subject-level (a limitation stated in the Conclusion). The never-released private test set is not touched. For lung sounds, where ICBHI publishes patient identifiers, the official 60/40 patient-independent (genuinely subject-level) split is adopted and then *repaired*: two patients whom the official file places on both sides of the split (identifiers 156 and 218) have all of their recordings forced to the training side, so that every patient lands on exactly one side. If the official split cannot be fetched and validated, the pipeline falls back to a seeded patient-level `GroupShuffleSplit` at the same 60/40 ratio. Either way the provenance of the split is logged.

Before any model is trained, a reusable assertion verifies that the training and test patient sets are disjoint:

```
def assert_no_patient_leakage(train_ids, test_ids):
    train, test = set(map(str, train_ids)), set(map(str, test_ids))
    overlap = train & test
    assert not overlap, f"PATIENT LEAKAGE: {len(overlap)} shared ids ..."
```

This check runs at the start of every experiment script, and its `[leakage-check OK] ... overlap=0` line is recorded in the experiment logs.

2.2 Preprocessing

All audio is loaded at its native sampling rate and resampled to a single common rate of 4000 Hz for both modalities, so that heart and lung sounds share one feature space (heart recordings are upsampled from their native 2000 Hz; lung recordings are downsampled from mixed native rates). At 4000 Hz the Nyquist limit of 2000 Hz comfortably covers the diagnostic bands of both modalities.

Each signal is then band-limited with a zero-phase Butterworth bandpass applied through second-order sections. The cutoffs are modality-specific: 20–400 Hz for heart sounds (retaining S1/S2 energy and murmurs while removing sub-cardiac drift) and 200–1800 Hz for lung sounds (the band of crackles and wheezes). The filter is fourth order and is applied with forward–backward filtering to preserve waveform timing; for segments too short for the forward–backward padding length the pipeline falls back to a single causal pass so that very short inputs remain finite and same-length. Finally each clip is peak-normalised to the range $[-1, 1]$; near-silent clips are left unchanged to avoid division blow-up. No global normaliser is fitted at this stage; feature-space standardisation is fitted on the training fold only (Section 2.5).

2.3 Segmentation

Heart recordings are sliced into fixed 3.0-second windows (12 000 samples at 4000 Hz). Training recordings use a 1.5-second hop (50% overlap) to increase the number of training windows, whereas test recordings use a 3.0-second hop (no overlap); this affects only the denominator of the recording-level majority vote, not correctness. A ragged final window is dropped, and a window in which more than 80% of samples are near-zero is discarded as silence. Respiratory sounds are not windowed but segmented at the *annotated cycle* boundaries supplied with ICBHI; each cycle is zero-padded (or trimmed) to a fixed 3.0-second length so that the downstream feature extractor sees a uniform input.

2.4 Feature extraction

Two parallel representations are produced from each fixed-length segment. Feature extraction is implemented with librosa 0.11.0 [21] for the classical path and torchaudio for the spectrogram path.

Classical features. MFCC extraction follows the standard mel-filterbank + DCT formulation [10]. From each segment we compute 40 mel-frequency cepstral coefficients together with their first- and second-order temporal deltas; each of these three blocks is summarised across frames by its mean and its standard deviation, yielding a 240-dimensional vector (feature set A). An extended set (set B, 250 -dimensional) appends the mean and standard deviation of five spectral statistics: spectral centroid, roll-off, bandwidth, zero-crossing rate and RMS energy. For respiratory cycles the segment is padded to 3.0 seconds *before* the cepstral transform; this is mandatory, because a raw short cycle yields too few analysis frames for the delta operator and would otherwise raise an error. Every emitted vector is asserted to have the expected dimension and to be free of non-finite values.

Deep-learning features. The same fixed 3.0-second segment is transformed into a 64×128 log-mel "image": a mel spectrogram (power spectrogram, 64 mel bands, 512-point FFT, hop 94) followed by amplitude-to-decibel conversion with an 80 dB dynamic range. The hop of 94 samples on a 12 000-sample window produces exactly 128 time frames. The mel band limits match the per-modality bandpass cutoffs (20–400 Hz for heart, 200–1800 Hz for lung); a 512-point FFT is used for both modalities to avoid an empty mel-filterbank warning that the narrow heart band triggers at smaller FFT sizes.

2.5 Models and training protocol

Classical models. Both feature sets are fed to four classifiers implemented in scikit-learn [22]: logistic regression, an RBF-kernel support-vector machine [11], a random forest [12] and gradient boosting (XGBoost [13]). Standardisation is fitted on the training fold only, inside a single scikit-learn pipeline, so no test-set statistics leak into training. Hyper-parameters are tuned by patient-grouped grid-search cross-validation for the two models where they materially affect performance (the SVM's C and γ, and XGBoost's tree depth and estimator count); logistic regression and the random forest are left at fixed, standard scikit-learn defaults. The classifier comparison therefore slightly favours the two tuned models, an asymmetry the reader should bear in mind. Class imbalance is handled by per-model class weighting applied inside the training fold only: balanced class weights for logistic regression, the SVM and the random forest, and a positive-class weight (binary heart) or balanced sample weights (four-class lung) for XGBoost. No synthetic oversampling is applied, which removes any risk of synthesising minority samples across the split.

Deep models. All deep models are implemented in PyTorch [23]. The compact convolutional network (SmallCNN) consists of four convolution–batch-normalisation–ReLU–max-pool blocks (baseline channel widths $1 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128$, treated as a tunable hyperparameter), an adaptive average pool, and a head with dropout (at least 0.3) before a linear classifier. The transfer-learning model is an EfficientNet-B0 backbone [16] pre-trained on ImageNet [24] (approximately 4.0 million parameters), with the single-channel log-mel image lifted to the three-channel 224×224 input the backbone expects via channel replication; an optional frozen-backbone mode trains only the classifier head as a CPU-feasible fallback. Deep models are trained with the Adam optimiser and a class-weighted cross-entropy loss, with early stopping on the validation primary metric, a fixed training-time cap, and best-checkpoint restoration; a train-versus-validation learning-curve figure is saved for each run to check for overfitting. Deep-model hyperparameters (learning rate, batch size, weight decay, dropout, augmentation strength, the class-imbalance sampling mode and, for the SmallCNN, the convolutional channel widths) were selected by a 128-trial bounded random search (32 trials per modality–model combination) that ranked configurations on the validation primary metric only, never on the test set. The reported heart SmallCNN therefore uses the wider $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channel variant selected by the search, whereas the lung SmallCNN retained the baseline widths; every final deep-learning result is reported as the mean ± standard deviation over three seeds $(1, 2, 42)$.

2.6 Evaluation

Both tasks are scored with the same balanced figure of merit, the arithmetic mean of sensitivity (Se) and specificity (Sp):

$$\mathrm{MAcc} = \frac{\mathrm{Se} + \mathrm{Sp}}{2}, \tag{1}$$

where sensitivity and specificity are the true-positive and true-negative rates of the confusion matrix,

$$\mathrm{Se} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{Sp} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}, \tag{2}$$

and TP, FN, TN, FP in Equation 2 are the true-positive, false-negative, true-negative and false-positive counts. Equation 1 weights the two classes equally irrespective of their prevalence, which is exactly why it is robust to the heavy class imbalance of both datasets and is the official metric of each challenge.

For heart sounds the primary metric is this *mean accuracy* (MAcc), with Se the recall on the abnormal class and Sp the recall on the normal class, the official CinC 2016 metric. Per-window predictions are first reduced to one prediction per recording by *majority vote*, because the recording is the clinically meaningful unit, and Equation 1 is then applied at the recording level; sensitivity, specificity, macro-F1, accuracy and, where a recording-level score is available, AUC-ROC are reported alongside.

For lung sounds the primary metric is the official *ICBHI score*. It uses the same balanced score (Equation 1), but Se is pooled over the three abnormal cycle classes (crackle, wheeze, both) while Sp is measured on the normal class; the score therefore captures normal-versus-abnormal discrimination rather than four-way accuracy. Per-class sensitivities and macro-F1 are reported in support.

Every model is additionally checked against a degenerate "predict-one-class" failure (predictions must occupy at least two confusion-matrix columns), and a confusion-matrix figure is saved for each run. Volumetric characteristics (sample counts per split, model parameter counts and training wall-clock times) are logged for the report's volumetric table (Annex B).

2.7 Reproducibility

A single configuration module is imported first by every script; its import side effect seeds Python's `random`, NumPy and PyTorch generators with the value 42 and sets cuDNN to deterministic mode. Some CUDA kernels nonetheless remain non-deterministic across hardware, so the deep-model results are reproduced as distributions over seeds rather than as bit-identical values. The software stack is pinned: Python 3.11, librosa 0.11.0, scikit-learn 1.8.0, XGBoost 3.2.0, PyTorch 2.11.0 with torchaudio 2.11.0, timm (≥ 1.0), imbalanced-learn 0.14.1, with the exact versions captured in a committed `requirements.txt`. The splits are written to disk and reloaded rather than regenerated, so that the partition is identical across runs. The full pinned environment is reproduced in Annex B. The classical pipelines run on CPU; the deep models (including the multi-seed, hyperparameter search and cross-modal runs) were trained on an NVIDIA A100 GPU, with each configuration kept small enough to remain CPU-feasible if required.

Two reproducibility caveats are stated plainly. First, all reported test metrics come from a *single* outer split per dataset (the seeded grouped split for heart, the official split for lung); the deep ±std values resample the seed on that fixed split, not the split itself, so no outer cross-validation is performed and split-selection variance is not quantified. Second, the Audio Spectrogram Transformer fine-tunes (Chapter 4) are bounded by a wall-clock budget rather than a fixed epoch count, so their epoch count — and hence the exact score — depends on host throughput; unlike the classical and epoch-pinned deep runs they are not bit-reproducible across hardware, and are reported as a seed mean ± standard deviation under that fixed time budget. Their producing script (`scripts/run_ast_multiseed.py`) is committed so the procedure, if not the exact number, can be re-run.

2.7.1 Chapter summary

The methodology is a single configuration-driven pipeline applied uniformly to both modalities, with leakage-safe grouped splits and an explicit zero-leakage assertion, modality-matched preprocessing and features, a common classical-versus-deep model set, and a shared balanced metric (Equation 1). This uniformity is what makes the cross-modal comparison of Chapter 4 meaningful.

3 Experimental results

This chapter reports the empirical results of the classical and deep-learning experiments on heart and lung sounds under the leakage-safe grouped-split protocol described in Chapter 2. All headline numbers are at the recording level for heart sounds and at the cycle level for lung sounds, using the official balanced metric of each task. Where a model predicted only one class the result is marked degenerate in the discussion; no such degenerate case arose in the final experiments. Deep-learning results are HPO-tuned (val-only selection, 128 trials) and reported as mean±std over three independent seeds $(1, 2, 42)$.

3.1 Unified comparison table

Table 3.1 collects every model trained in this study under a single schema, enabling direct cross-method and cross-modality comparison. All 20 rows are final. Rows are grouped first by modality, then by feature set, then by model. Within each modality the best primary-metric score is shown in bold. Deep-learning scores represent HPO-tuned mean±std over three seeds.

Table 3.1 – Unified model comparison across both modalities. Heart metric: MAcc = (Se + Sp)/2 (CinC 2016 official). Lung metric: ICBHI Score = (Se + Sp)/2 (ICBHI 2017 official). DL rows report HPO-tuned mean±std over three seeds $(1, 2, 42)$, val-only HPO selection (128 trials). Feature sets: A = MFCC+Δ+ΔΔ (240-d); B = MFCC+Δ+ΔΔ+spectral (250-d); log-mel = 64×128 image.

Modality	Feature set	Model	Metric	Score	Se	Sp	macro-F1
Heart	A (MFCC+Δ)	LogReg	MAcc	0.794	0.869	0.720	0.706
Heart	A (MFCC+Δ)	SVM	MAcc	0.859	0.915	0.802	0.783
Heart	A (MFCC+Δ)	RF	MAcc	0.817	0.692	0.942	0.827
Heart	A (MFCC+Δ)	XGBoost	MAcc	0.879	0.915	0.843	0.816
Heart	B (MFCC+Δ+spec)	LogReg	MAcc	0.825	0.908	0.742	0.734
Heart	B (MFCC+Δ+spec)	SVM	MAcc	0.869	0.938	0.800	0.788
Heart	B (MFCC+Δ+spec)	RF	MAcc	0.831	0.723	0.940	0.837
Heart	**B (MFCC+Δ+spec)**	**XGBoost**	**MAcc**	**0.903**	**0.946**	**0.859**	**0.839**
Heart	log-mel 64×128	SmallCNN	MAcc	0.871 ± 0.009	0.910	0.831	—
Heart	log-mel 64×128	EfficientNet-B0	MAcc	0.898 ± 0.008	0.936	0.860	—
Lung	A (MFCC+Δ)	LogReg	ICBHI	0.507	0.656	0.358	0.273
Lung	A (MFCC+Δ)	SVM	ICBHI	0.536	0.618	0.454	0.319
Lung	A (MFCC+Δ)	RF	ICBHI	0.476	0.101	0.851	0.226
Lung	A (MFCC+Δ)	XGBoost	ICBHI	0.511	0.470	0.551	0.297
Lung	B (MFCC+Δ+spec)	LogReg	ICBHI	0.533	0.693	0.372	0.291
Lung	B (MFCC+Δ+spec)	SVM	ICBHI	0.537	0.615	0.458	0.320
Lung	B (MFCC+Δ+spec)	RF	ICBHI	0.472	0.098	0.847	0.243
Lung	B (MFCC+Δ+spec)	XGBoost	ICBHI	0.500	0.369	0.631	0.294
Lung	log-mel 64×128	SmallCNN	ICBHI	0.540 ± 0.022	0.755	0.325	—
Lung	**log-mel 64×128**	**EfficientNet-B0**	**ICBHI**	**0.555 ± 0.016**	**0.509**	**0.601**	**—**

DL rows: HPO-tuned (128-trial bounded random search, val-only selection) mean±std over seeds $\{1, 2, 42\}$. Macro-F1 not computed for multi-seed DL (indicated by —); macro-F1 figures for single-seed runs appear in Annex A. The fine-tuned Audio Spectrogram Transformer (Chapter 4) reaches a higher lung score still (ICBHI = 0.594 ± 0.023) but is excluded from this table because it was trained under a non-comparable budget (no HPO, AudioSet pretraining, few epochs); EfficientNet-B0 is the best model in the controlled comparison.

3.2 Heart-sound results (PhysioNet/CinC 2016)

3.2.1 Classical models

All eight classical configurations (two feature sets × four classifiers) were evaluated on the same held-out test set of 626 recordings (split at the recording level; see Chapter 2). The best classical result was achieved by XGBoost on feature set B (MFCC+Δ+ΔΔ + spectral statistics), reaching MAcc = 0.903, with Se = 0.946 and Sp = 0.859. This result confirms the practical value of the five additional spectral statistics: XGBoost on feature set A scored MAcc = 0.879, so the richer feature set adds a meaningful 2.3 percentage points.

This figure is not directly comparable to the CinC 2016 challenge leaderboard (top entries ≈ 0.86; Chapter 1): that ranking was computed on the withheld private test set, whereas our value comes from a self-constructed recording-level partition of the public training pool. Because the heart split is recording-level rather than strictly subject-level (Section 2.1), the absolute score may be optimistic relative to a fully subject-disjoint evaluation; the contribution is the controlled cross-method comparison under one fixed protocol, not the absolute number itself.

SVM was the second-best classical model across both feature sets (MAcc 0.859 on A, 0.869 on B). This ordering is statistically robust, not a rounding artefact: a McNemar paired test on the 626 recording-level predictions finds XGBoost-B and SVM-B disagree on 48 recordings, of which XGBoost is correct on 39 and SVM on only 9 ($\chi^2 = 17.5$, exact $p \approx 1.5 \times 10^{-5}$). SVM and XGBoost both achieve high sensitivity (Se ≥ 0.915 in all configurations), reflecting the fact that class-weighted training forces these models to capture true positives. By contrast, random forest reaches high specificity (Sp ≥ 0.940) but lower sensitivity (Se ≈ 0.69–0.72), suggesting that RF's ensemble structure defaults more readily to the majority normal class. Logistic regression, despite being the simplest linear model, achieves a respectable MAcc of 0.825 on feature set B, which indicates that the MFCC+spectral feature space is well-structured for linear separation.

Figure 3.1 shows the confusion matrix for the best model (XGBoost, set B). Out of 130 abnormal test recordings, 123 are correctly identified (7 false negatives), and out of 496 normal recordings, 426 are correctly identified (70 false positives), consistent with the reported Se = 0.946 and Sp = 0.859. For a screening or triage tool this is a favourable operating point: at Se = 0.946 only about 5% of abnormal recordings are missed (the error that matters most when the aim is to flag cases for expert review), while the false positives implied by Sp = 0.859 would be cleared by a confirmatory examination.

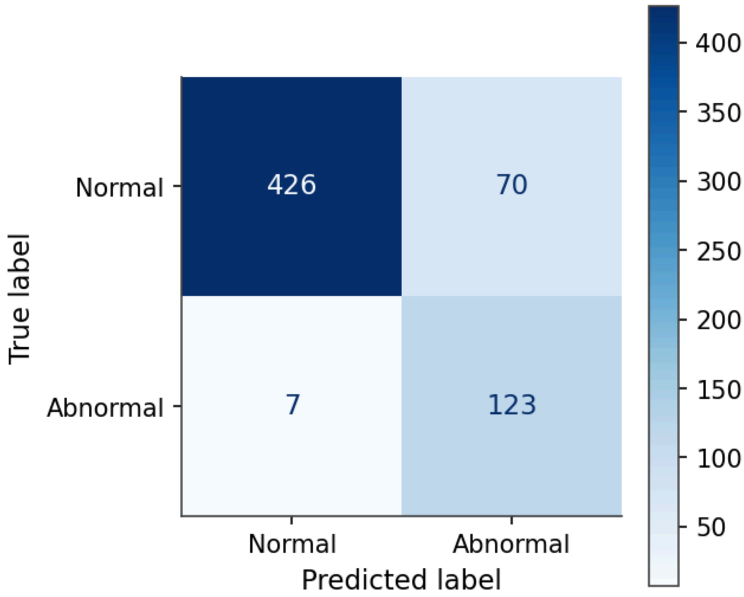

Figure 3.1 – Confusion matrix of the best heart-sound classical classifier (XGBoost, feature set B: MFCC+Δ+ΔΔ + spectral statistics). Rows: true class; columns: predicted class. Test set: 626 heart recordings (recording-level split; CinC 2016 publishes no recording-to-subject map, see Section 2.1).

For comparison, Figure 3.2 shows the SVM confusion matrix (feature set B). The SVM achieves similar sensitivity but with slightly fewer false positives, reflecting a different decision-threshold behaviour.

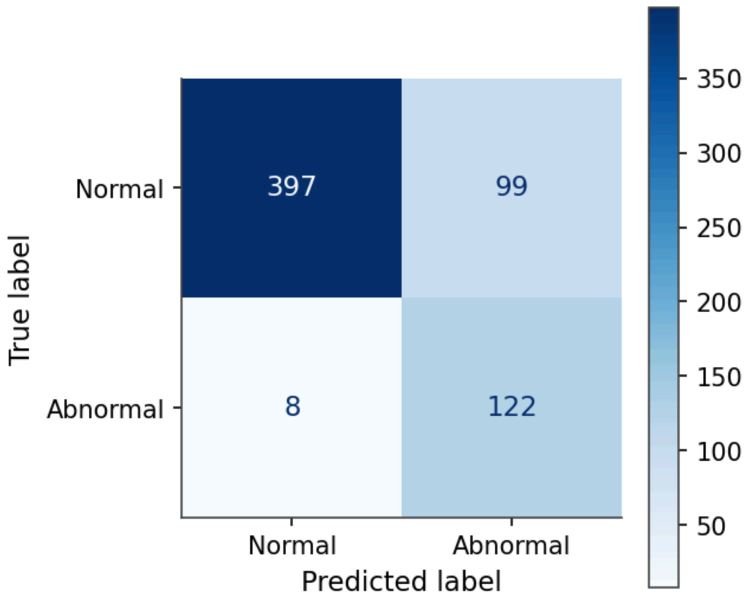

Figure 3.2 – Confusion matrix of the second-best heart-sound classical classifier (SVM, feature set B). Same test set as Figure 3.1.

3.2.2 Deep-learning models

The deep-learning models were evaluated with HPO-tuned hyperparameters (128-trial bounded random search, validation-only selection) and results are reported as mean±std over three independent seeds. The compact SmallCNN trained on 64×128 log-mel spectrograms reached MAcc = 0.871 ±

0.009 ($\mathrm{Se} = 0.910, \mathrm{Sp} = 0.831$), and EfficientNet-B0 reached $\mathrm{MAcc} = 0.898 \pm 0.008$ ($\mathrm{Se} = 0.936, \mathrm{Sp} = 0.860$). With hyperparameter tuning, EfficientNet-B0 reaches a mean accuracy comparable with the best classical model (XGBoost-B, $\mathrm{MAcc} = 0.903$): the difference of 0.005 is smaller than EfficientNet-B0′s own seed-to-seed standard deviation (± 0.008), so the two models are of comparable accuracy on this task, with the classical model remaining numerically best. The comparison can be made more precise: a normal-approximation 95% confidence interval on the recording-level MAcc of the best classical model, computed from its per-class binomial counts ($\mathrm{Se} = 123/130$, $\mathrm{Sp} = 426/496$), is approximately $[0.878, 0.927]$. EfficientNet-B0′s mean (0.898) lies inside this interval. We can go further than interval overlap with a directly paired test: running the retained in-domain EfficientNet-B0 checkpoints (the cross-modal reference models of Chapter 4, per-seed MAcc 0.878/0.885/0.898) back through recording-level inference gives the deep model's per-recording predictions, and a McNemar test pairs them against XGBoost-B on the same 626 recordings. (Because the EfficientNet-B0 checkpoints are too large to ship, these per-seed prediction vectors are committed to the repository as a fixed CSV, so the paired test is reproducible while the upstream inference is not re-run from scratch.) The verdict tracks the seed. On the two seeds where EfficientNet-B0 lands at 0.878 and 0.885 it is significantly behind XGBoost-B ($\chi^2 = 15.6$, $p \approx 10^{-4}$; and $\chi^2 = 6.4$, $p = 0.011$), but on its strongest seed, where it reaches 0.898, the two models are statistically indistinguishable (discordant 25 vs 29 recordings, $\chi^2 = 0.17$, $p = 0.68$). The classical-versus-deep ordering is therefore unstable across seeds, which is precisely the within-seed noise the headline comparison reports: a single-seed McNemar test could have shown either that XGBoost dominates or that the two are equal, depending on the seed drawn. This is the central justification for the multi-seed protocol used throughout. It also reframes the earlier core-run claim that classical methods dominated: after equivalent HPO, the two families are of comparable accuracy on heart sounds.

The loss curves for both deep models (Figure 3.3 and Figure 3.4) show the training loss falling steadily; the validation loss trends downward for SmallCNN and plateaus for EfficientNet-B0, with no runaway overfitting. Early stopping restores the best validation checkpoint, confirming that the dropout (≥ 0.3) and early-stopping regularisation are effective.

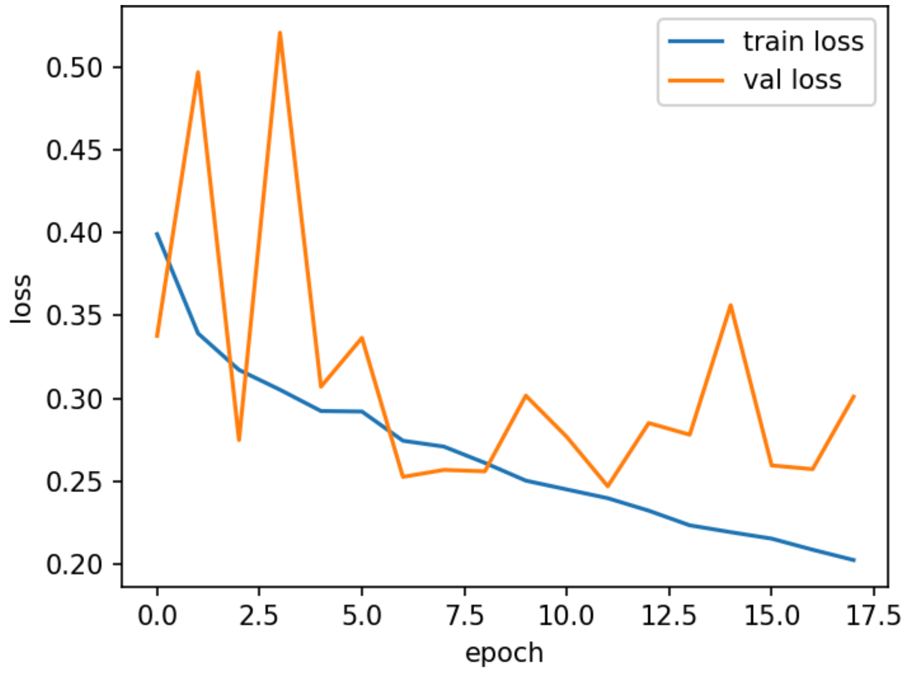

Figure 3.3 – Training and validation loss per epoch for SmallCNN on heart sounds. The training loss falls steadily and the (noisier) validation loss trends downward, indicating convergence without runaway overfitting.

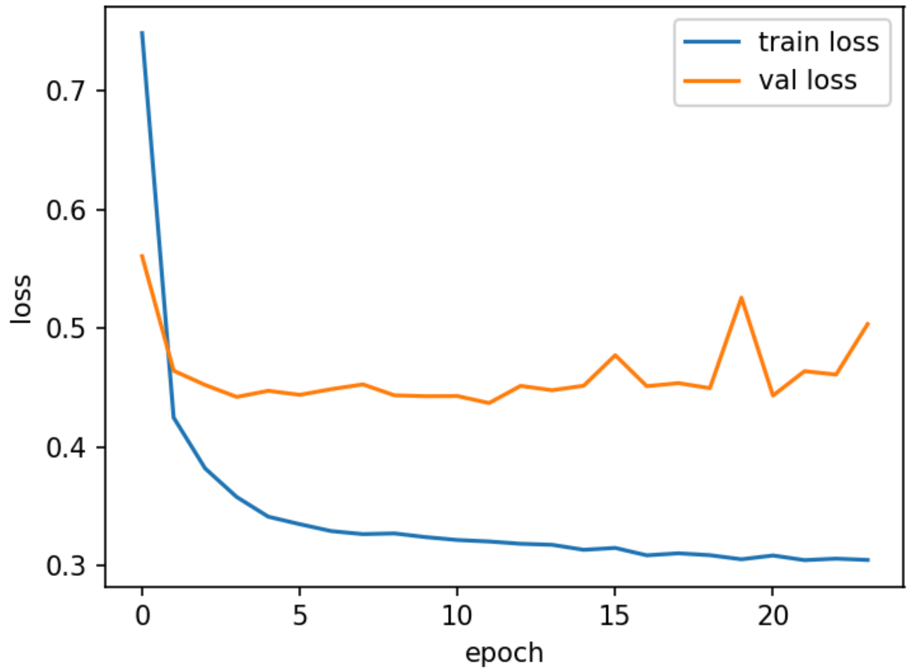

Figure 3.4 – Training and validation loss per epoch for EfficientNet-B0 on heart sounds. The validation loss plateaus while the training loss keeps falling, the modest train–validation gap typical of fine-tuning; early stopping selects the best checkpoint.

Figure 3.5 shows the confusion matrix of EfficientNet-B0, the strongest deep model on heart sounds. Its recording-level error pattern mirrors the best classical model: most abnormal recordings are detected, with only a modest number of false positives on the normal class.

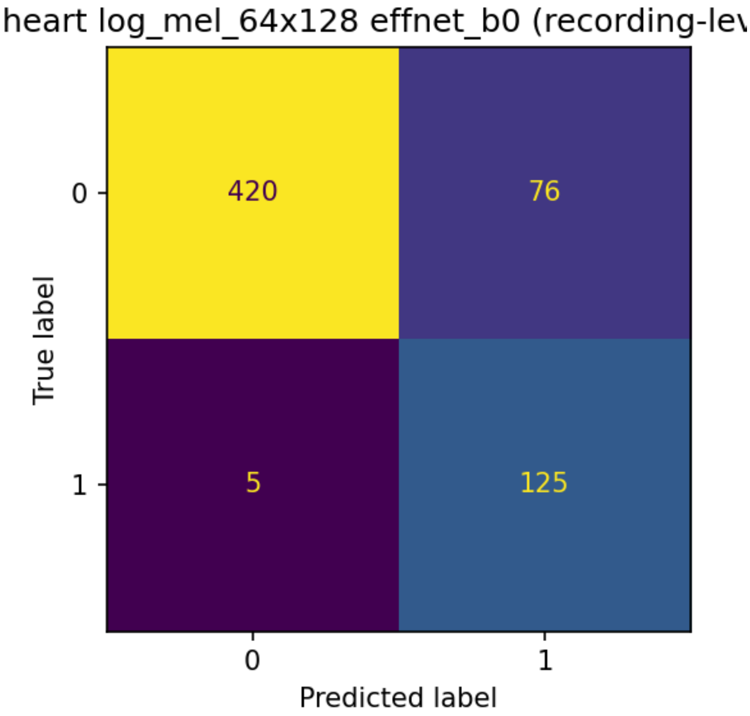

Figure 3.5 – Confusion matrix of the best heart-sound deep model (EfficientNet-B0, log-mel spectrograms; single representative run). Rows: true class; columns: predicted class. Test set: 626 heart recordings.

3.3 Lung-sound results (ICBHI 2017)

3.3.1 Classical models

All eight classical configurations were evaluated on the official ICBHI 60/40 patient-independent test set of 2636 cycles from 47 test patients. ICBHI scores are markedly lower than heart MAcc across all models, reflecting the fundamentally harder nature of this task: four imbalanced classes (three minority abnormal types) versus two classes.

The best classical result is SVM on feature set B with ICBHI score = 0.537 (Se = 0.615, Sp = 0.458). The score on feature set A is nearly identical (0.536), suggesting that the spectral statistics add marginal value for lung classification compared to the clear boost they provided for heart classification. XGBoost on feature set A scores 0.511 and declines on feature set B (0.500), which is the opposite pattern from heart sounds, an observation directly relevant to the cross-modal analysis of Chapter 4.

A striking pattern in the lung results is the contrast between random forest and all other classifiers. RF achieves very high specificity (Sp ≈ 0.85) but extremely low sensitivity on the abnormal classes (Se ≈ 0.10). This "majority-class collapse" (predicting almost exclusively normal) is a failure mode specific to RF's averaging behaviour when classes are highly imbalanced and class weighting alone has not fully counteracted the imbalance within the training fold. No separate figure is shown for the random forest; its collapse is evident directly in the per-class sensitivities of Table 3.1 (abnormal-class Se ≈ 0.10). For contrast, Figure 3.6 shows the best classical model, the SVM, whose predictions are distributed across all four classes rather than collapsing to the majority.

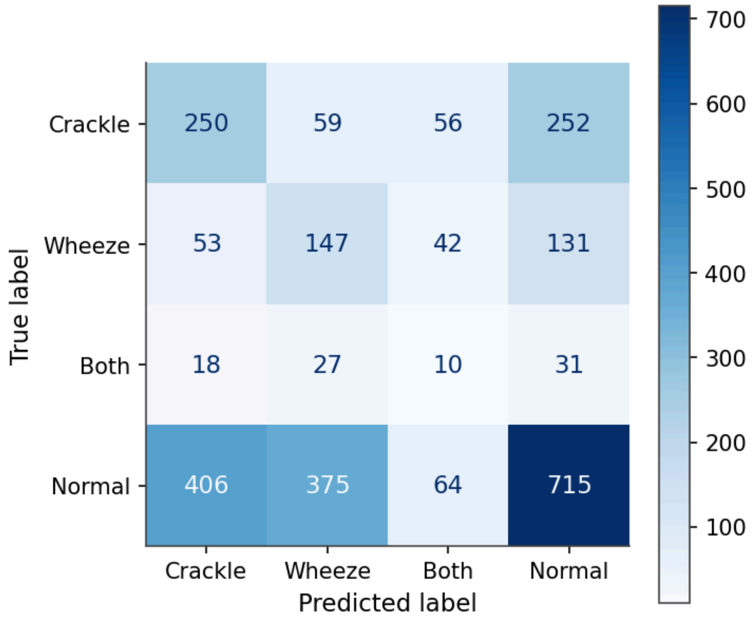

Figure 3.6 – Confusion matrix of the best lung-sound classical classifier (SVM, feature set B). Rows: true class; columns: predicted class. Test set: 2636 cycles from 47 patients.

Logistic regression on feature set B achieves ICBHI = 0.533, comparable to SVM. Both linear models benefit from class-weighted loss, which forces predictions across all four classes rather than collapsing to the majority.

The per-class sensitivities in Annex A locate the difficulty. The rare combined crackle-and-wheeze class is barely recovered by any classifier: its per-class sensitivity runs from about 0.11 to 0.16 for the linear and boosting models and falls to almost zero for random forest. Crackle and wheeze are captured only partially, with sensitivities of roughly 0.2 to 0.4. Because the ICBHI score pools sensitivity over these three abnormal classes, it is the scarcity of the minority classes, not a weakness of any single algorithm, that holds every classical method near 0.54. The deep models meet the same bottleneck, which is why their larger capacity buys so little on this task. Clinically the consequence is concrete: a tool built on this data would reliably separate normal from abnormal breathing but could not yet be trusted to name which adventitious sound is present.

3.3.2 Deep-learning models

The deep-learning lung models are also reported as HPO-tuned mean±std over three seeds. The SmallCNN reached ICBHI = 0.540 ± 0.022 (Se = 0.755, Sp = 0.325), and EfficientNet-B0 reached ICBHI = 0.555 ± 0.016 (Se = 0.509, Sp = 0.601). Note that the multi-seed mean for CNN (0.540) is slightly below an earlier single-seed run (0.551 reported at the core-run stage); this honest regression reflects seed variance on the harder four-class task rather than a model change. The EfficientNet-B0 is the best lung model in the main comparison (0.555), marginally ahead of SmallCNN and the best classical model (SVM-B, 0.537); the supplementary AST of Chapter 4 scores higher still (0.594 ± 0.023) but sits outside this controlled comparison and is reported separately. The deep-vs-classical difference on lung is small (roughly 1.5–2 pp) and within the multi-seed standard deviation, placing classical and deep methods in the same performance tier on this task. Both deep models show better sensitivity on minority abnormal classes compared to random forest, which is reflected in a more balanced confusion matrix (Figure 3.7).

This best ICBHI score (0.555) sits below the strongest published results on the official split (patch-mix AST and related transformer systems reach the low 0.60s), and the gap reflects scope rather than a methodological flaw: those systems use transformer backbones with large-scale audio pretraining and heavy augmentation, whereas our deep models are a compact CNN and an ImageNet-pretrained EfficientNet-B0 trained under a deliberately modest, leakage-controlled protocol. As Chapter 1 notes, a number of higher reported ICBHI scores additionally rely on non-official or recording-level splits that are not comparable to the official patient-independent partition adopted here.

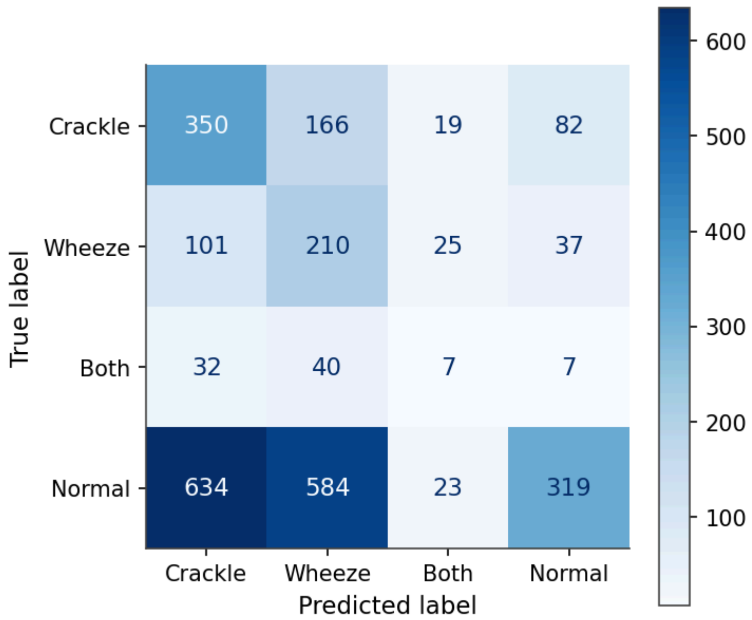

Figure 3.7 – Confusion matrix of the lung-sound SmallCNN classifier (on log-mel spectrograms). The log-mel representation enables better discrimination of crackle and wheeze relative to classical models.

The best lung model in the main comparison, EfficientNet-B0, is shown in Figure 3.8. Relative to the SmallCNN it shifts the operating point toward specificity at the cost of sensitivity, but both deep models distribute their predictions across all four classes rather than collapsing to the majority normal class as random forest does. Note that the headline ICBHI score is a *binary* normal-versus-abnormal measure (sensitivity pooled over the three abnormal classes); genuine four-way discrimination, visible in the per-class sensitivities of Annex A and weakest for the scarce “both” class, is substantially lower than these pooled figures suggest.

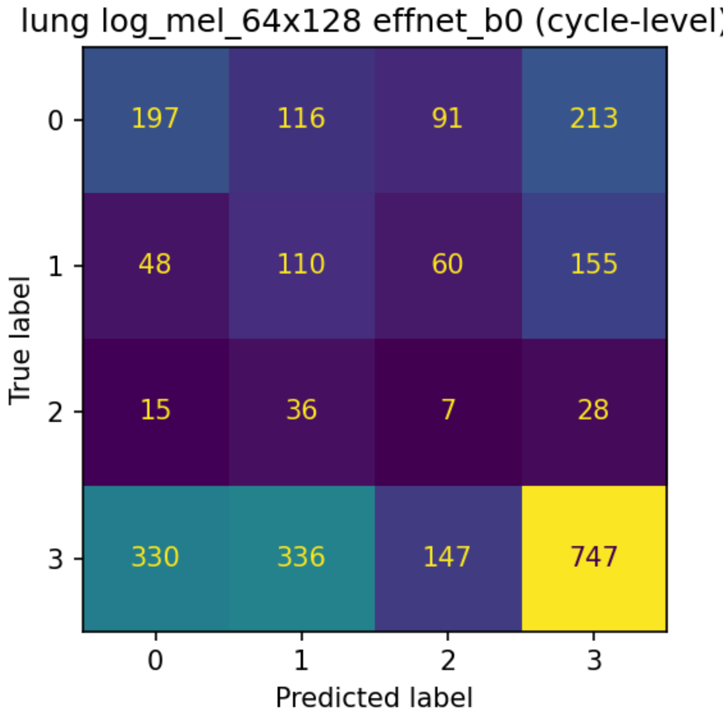

Figure 3.8 – Confusion matrix of the best lung model in the main comparison (EfficientNet-B0, log-mel spectrograms; single representative run). Test set: 2636 cycles from 47 patients.

Figure 3.9 shows the loss curve for the lung CNN. The training loss falls while the validation loss flattens and then begins to climb after roughly epoch 9 (the over-fitting expected on this harder four-class task) so early stopping restores the earlier best checkpoint.

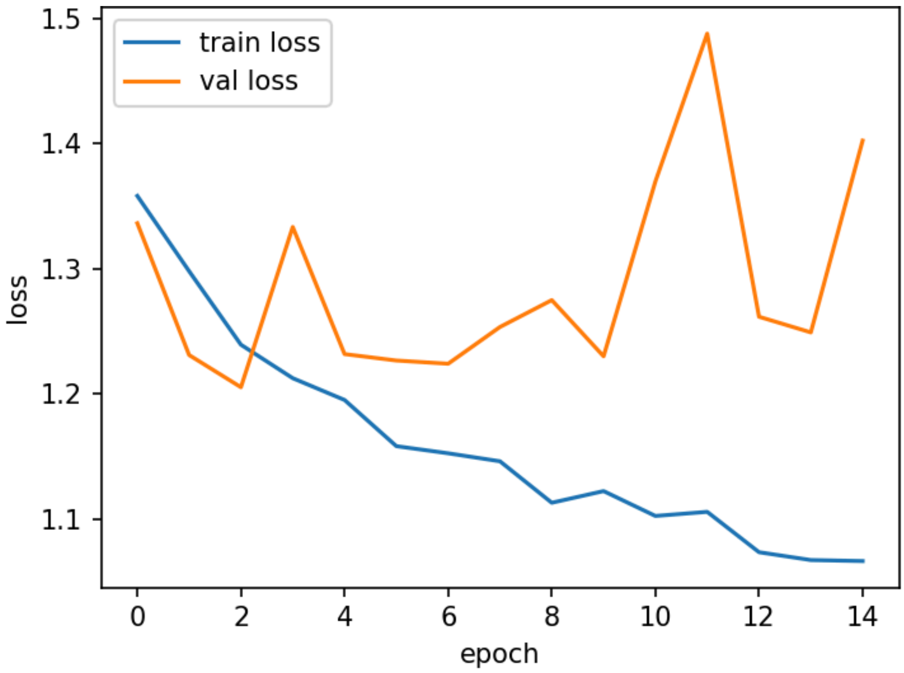

Figure 3.9 – Training and validation loss per epoch for SmallCNN on lung sounds. The validation loss starts rising after roughly epoch 9 (over-fitting on the harder four-class task), so early stopping restores the earlier best checkpoint.

3.4 Heart-murmur detection on CirCor DigiScope 2022 (extension)

The binary normal/abnormal task on CinC 2016 does not localise or grade a specific pathology. To test whether the same pipeline extends to a clinically richer heart task, and on a genuinely patient-level split, we applied it to the CirCor DigiScope 2022 phonocardiogram dataset [9], the largest public pediatric PCG corpus. CirCor provides patient-level murmur annotations (Present / Absent / Unknown); we keep the 874 subjects carrying a definite label, drop the "Unknown" cases, and form a binary murmur-detection task. Because CirCor records several auscultation locations per subject under one patient identifier, the split is made strictly at the patient level (611 train / 263 test subjects; 3 007 recordings, 912 in the test set), removing the subject-level ambiguity that the CinC heart split could not rule out (Section 2.1). All preprocessing, windowing, feature extraction and models are reused unchanged; scores are recording-level MAcc, deep models as mean±std over the same three seeds $(1, 2, 42)$.

Table 3.2 reports the outcome, and its headline is a reversal of the CinC ordering. On murmur detection the deep SmallCNN is clearly best ($\text{MAcc} = 0.810 \pm 0.007$, $\text{Se} = 0.71$, $\text{Sp} = 0.91$), ahead of EfficientNet-B0 (0.760 ± 0.018) and well clear of every classical model, whose best is logistic regression on feature set B ($\text{MAcc} = 0.688$). Whereas on the easier CinC normal/abnormal screen the tuned classical XGBoost narrowly led the deep models, on the subtler murmur task the convolutional model opens a gap of about 0.12 over the strongest classical baseline. Two factors plausibly explain the shift: a murmur is a localised spectro-temporal event that a learned spectrogram filter captures better than pooled MFCC statistics, and the larger CirCor training set (29 142 windows) gives the deep models more to learn from. The random forest again collapses toward the majority "Absent" class ($\text{Se} \approx 0.04$–0.10), the same imbalance failure mode seen on lung sounds.

This extension supports two claims made elsewhere in the report. It shows that the pipeline transfers to a third heart dataset and a more clinically meaningful task with only a configuration change, and it demonstrates that the classical-versus-deep verdict is task-dependent: classical methods match or beat deep learning on the coarse normal/abnormal screen, but deep learning is the stronger choice once the task requires resolving a specific acoustic sign such as a murmur. The advantage here is carried by the SmallCNN, which outscores the larger EfficientNet-B0 (the reverse of their CinC ordering); since the deep configurations were not given a CirCor-specific hyperparameter search, the claim is that *a* deep model, not deep learning categorically, wins this task. Because CirCor's split is patient-level, this is also the study's cleanest single piece of evidence that the deep pipeline generalises beyond window-level memorisation.

Table 3.2 – Heart-murmur detection on CirCor DigiScope 2022 (binary Present vs. Absent, patient-level split, 912 test recordings). Recording-level MAcc = (Se + Sp)/2; deep models as mean±std over seeds $\{1, 2, 42\}$. Classical rows give the best configuration of each classifier across feature sets A/B.

Model	MAcc	Se	Sp	Family
SmallCNN (log-mel)	**0.810 ± 0.007**	**0.71**	**0.91**	**deep**
EfficientNet-B0 (log-mel)	0.760 ± 0.018	0.59	0.93	deep
LogReg (set B)	0.688	0.65	0.73	classical
XGBoost (set B)	0.685	0.56	0.81	classical
SVM (set B)	0.668	0.50	0.83	classical
Random forest (set B)	0.548	0.10	1.00	classical

3.5 Volumetric characteristics

Table 3.3 collects the dataset sizes, model parameter counts and training times that characterise the experimental scale. These are required by Annex 5 §2.5. In total the study evaluates 20 distinct model configurations under one protocol (4 classical algorithms $\times$ 2 feature sets $+2$ deep models, on each of the two primary modalities), with a further ten on the CirCor murmur extension; each deep configuration is additionally the outcome of a 128-trial hyperparameter search and is reported as a mean over 3 seeds, so the deep results alone rest on several hundred individual training runs. Beyond the data and model sizes, the experimental pipeline itself comprises 6 619 lines of Python across 41 modules in `src/` and `scripts/` (approximately 236 KB of source code), supported by a further 2 093 lines of automated tests. Full volumetric details appear in Annex B, and per-class cycle counts in Annex C.

Table 3.3 – Volumetric characteristics of the experiments. Classical training times are CPU wall-clock seconds; deep-learning training times are GPU (NVIDIA A100) wall-clock seconds for the HPO-selected configuration. Heart patient counts are not reported: CinC 2016 provides no recording-to-subject map, so the heart split is grouped at the recording level (Section 2.1).

Quantity	Heart	Lung
Train segments/cycles	33 246	4 262
Test segments/cycles	4 167	2 636
Train recordings / patients	2 500 / —	551 / 79
Test recordings / patients	626 / —	369 / 47
Best classical model (SVM-B) train time (s)	659	31
Best classical model (XGB-B) train time (s)	21	26
SmallCNN parameters (HPO-tuned)	389 314	98 148
SmallCNN train time (s), GPU	262	26
EfficientNet-B0 parameters	4 010 110	4 012 672
EfficientNet-B0 train time (s), GPU	1 345	204
Log-mel data volume (MB)	1 227	226
Classical feature data volume (MB)	74	14

3.5.1 Chapter summary

On heart sounds, the best overall result is XGBoost with MFCC+Δ+$\Delta\Delta$ + spectral statistics (feature set B), reaching MAcc = 0.903 (Se = 0.946, Sp = 0.859). After HPO tuning, the deep EfficientNet-B0 reaches MAcc = 0.898 ± 0.008, comparable with the best classical model (the 0.005 difference is below the model's ±0.008 seed standard deviation), though the classical model remains numerically best. SmallCNN reaches MAcc = 0.871 ± 0.009. After equivalent tuning effort the two method families are of comparable accuracy on heart sounds, in contrast to the larger classical advantage seen in the untuned core run.

On lung sounds, the best result across all 20 configurations is EfficientNet-B0 with ICBHI = 0.555 ± 0.016, narrowly ahead of the best classical model (SVM-B, 0.537) and SmallCNN (0.540 ± 0.022). Classical and deep models occupy the same performance tier on this task; the four-class imbalanced structure limits all methods equally. The gap between heart and lung scores (MAcc ≈ 0.90 vs. ICBHI ≈ 0.55) reflects a real difficulty difference — a binary task on longer recordings versus a four-class imbalanced task on short cycles — but the two metrics are not on a common scale (binary MAcc versus pooled-abnormal ICBHI), so the gap is partly metric-structural and should not be read as a pure difficulty ratio. Chapter 4 analyses what these differences imply for cross-modal transfer of method rankings.

4 Cross-modal analysis and arterial sub-study

This chapter develops the project's primary novel contribution: an analysis of how method families transfer across auscultation modalities, combining the per-modality Chapter 3 results with dedicated cross-modal transfer experiments (pretrained encoder applied to the opposite modality) and a joint multi-task probe. It then provides an analytical treatment of arterial sounds, for which no open dataset exists, documenting the acoustic basis, the data-availability obstacle, and how the pipeline would extend if the obstacle were removed.

4.1 Transfer of method rankings across modalities

Because heart and lung models were evaluated under one shared protocol and one metric family (the same preprocessing, features, model set and balanced score; only the grouping granularity and test fraction differ, as the datasets dictate), their rankings can be compared directly. We examine the question in three ways: (i) the ordinal ranking of classifiers within each modality, (ii) the relative advantage of feature set B over feature set A, and (iii) the relative standing of classical versus deep models.

4.1.1 Classifier ranking

For heart sounds the ranking (best to worst) by primary metric is: XGBoost-B (0.903) > XGBoost-A (0.879) > SVM-B (0.869) > SVM-A (0.859) > RF-B (0.831) > LogReg-B (0.825) > RF-A (0.817) > LogReg-A (0.794). Collapsing per feature set, the ordering at the classifier level is XGBoost > SVM > RF > LogReg.

For lung sounds the ranking by ICBHI score is substantially different: SVM-B (0.537) ≈ SVM-A (0.536) > LogReg-B (0.533) > XGBoost-A (0.511) > LogReg-A (0.507) > XGBoost-B (0.500) > RF-A (0.476) ≈ RF-B (0.472). The lung classifier ordering is SVM ≈ LogReg > XGBoost > RF.

Two inversions are notable. First, XGBoost, the dominant model on heart sounds, falls to third place on lung sounds and actually degrades from feature set A to B (0.511 to 0.500), the reverse of its heart behaviour. Second, logistic regression rises to near-parity with SVM on lung sounds, suggesting that the four-class MFCC feature space is more linearly separable than the two-class heart space, or that gradient boosting's additional capacity introduces over-fitting on the smaller lung training set (4262 cycles versus 33 246 heart windows). These rank inversions are the central qualitative observation of the cross-modal analysis; the rank correlation below is computed first on these four classifiers and then re-checked on an expanded nine-classifier panel (Table 4.1) to confirm it is not an artefact of a small sample.

A prediction-level paired test sharpens the same point and removes any suspicion that the inversion is a rounding effect. On heart, a McNemar test on the 626 recording-level predictions confirms the rank-1 model is genuinely ahead of the rank-2: XGBoost-B beats SVM-B on 39 of their 48 discordant recordings ($\chi^2 = 17.5$, $p \approx 1.5 \times 10^{-5}$). On lung the very same pair is reordered, and instructively so. At the raw cycle level XGBoost-B is in fact *significantly more accurate* than SVM-B (it wins 401 of their 711 discordant cycles, $\chi^2 = 11.4$, $p \approx 7 \times 10^{-4}$), yet SVM-B attains the higher balanced ICBHI score

(0.537 vs. 0.500), because XGBoost buys its raw-accuracy edge by leaning on the majority normal class while the balanced metric refuses to reward that. The model that wins on heart therefore does not win on lung under the metric that matters, and the reason is visible in the individual predictions, not merely in rounded aggregates, exactly the non-transfer of rankings this section documents.

4.1.2 Feature set effect

On heart sounds, adding spectral statistics (feature set B) consistently improves every classifier, with the largest absolute gain for XGBoost (+2.3 pp) and modest gains for SVM (+1.1 pp) and LogReg (+3.0 pp). On lung sounds, the feature-set effect is inconsistent: SVM improves negligibly (+0.1 pp), LogReg improves (+2.6 pp), but XGBoost worsens (−1.1 pp) and RF also worsens. The spectral statistics that capture cardiac sound "brightness" (centroid, roll-off) are apparently less informative for the respiratory task, where the shape of adventitious sound transients is more relevant than broadband spectral shape.

4.1.3 Classical versus deep learning

With HPO-tuned hyperparameters and three-seed averaging, the picture on heart sounds is more nuanced than the core run suggested. EfficientNet-B0 reaches MAcc $= 0.898 \pm 0.008$ (Se $= 0.936$, Sp $= 0.860$), comparable with the best classical model (XGBoost-B, MAcc $= 0.903$): the 0.005 difference is below EfficientNet-B0′s ± 0.008 seed standard deviation, though the classical model remains numerically best. SmallCNN reaches MAcc $= 0.871 \pm 0.009$. The clear classical advantage seen in the untuned core run thus shrinks to a margin within seed noise after equivalent optimisation effort, without being reversed.

On lung sounds, the deep models achieve ICBHI $= 0.540 \pm 0.022$ (CNN) and 0.555 ± 0.016 (EfficientNet-B0), placing them marginally ahead of but practically indistinguishable from the best classical model (SVM-B, 0.537). Both modalities therefore converge on the same finding: with proper HPO, classical and deep methods occupy the same performance tier on these tasks and dataset scales.

4.1.4 Summary of cross-modal findings (classical)

A Spearman rank correlation [25] between the per-classifier ICBHI scores (lung) and the corresponding MAcc scores (heart), computed over only the four classical classifier types, is statistically uninformative: $\rho = 0.60$ on feature set A but $\rho = 0.00$ on feature set B ($n = 4$ each), and pooling both sets gives $\rho = 0.24$ ($n = 8$), none significant at conventional thresholds. At this sample size the test can reject neither "no correlation" nor "perfect correlation", so we report these coefficients only for completeness and do *not* rest the non-transfer claim on them. The claim is carried instead by the qualitative rank inversion documented above and by the prediction-level McNemar test: XGBoost's dominance on heart (rank 1) does not carry over to lung (it falls to third, and even reverses its feature-set preference), whereas SVM is the one model competitive on both, and the paired test confirms that the very same XGBoost–SVM pair reorders between modalities under the balanced metric. The practical implication is that a researcher who tuned a model exclusively on heart data and redeployed it on lung data would incur a non-trivial performance penalty, which is precisely the argument for per-modality model selection even within a shared pipeline.

4.1.5 Robustness: a nine-classifier panel

A correlation over four classifiers is a thin estimate, so we tested whether the non-transfer of rankings survives a larger panel. Five further class-weighted classifiers were added under the identical protocol, Extra-Trees, AdaBoost, a ridge linear classifier, an SGD-trained logistic model and a linear-kernel SVM, and all nine were re-evaluated on both modalities (Table 4.1). The five additions use fixed defaults, like the untuned logistic-regression and random-forest baselines, so the panel stays internally consistent.

The expansion supports the finding through the ranks themselves, not through the correlation coefficient, which stays underpowered even at $n = 9$ (heart-versus-lung Spearman $\rho = 0.42$, $p = 0.26$ on set A; $\rho = 0.13$, $p = 0.73$ on set B; pooled $\rho = 0.32$, $n = 18$, $p = 0.19$, none significant). What is informative is the ordering, and the headline inversion grows sharper: XGBoost is rank 1 of 9 on heart but falls to rank 7 of 9 on lung, while the radial-basis SVM is the only classifier in the top two of both. More informative is a structure the four-model panel could not expose. The tree ensembles (XGBoost, AdaBoost, random forest, Extra-Trees) occupy the upper half on heart but the lower half on lung, whereas the linear models (ridge, logistic regression, SGD-logistic, linear SVM) do the opposite, clustering near the top on lung. The non-transfer of rankings is therefore not an artefact of a small panel but a model-family-by-modality interaction: non-linear tree ensembles suit the binary heart task, linear models are relatively stronger on the four-class lung task, and only the kernel SVM is robust to both. This is a concrete, quantified reason to select the classifier per modality even inside one shared pipeline. The caveat is that seven of the nine classifiers (all but the tuned SVM and XGBoost) run at library defaults, so the precise mid-panel ranks could shift under per-model tuning; the claim is therefore a property of the pipeline as configured, not an architecture-intrinsic law, though the top/bottom split by model family is too large to be a pure tuning artefact.

Table 4.1 – Nine-classifier panel under the shared protocol (feature set B). Heart score is recording-level MAcc; lung score is the cycle-level ICBHI Score; ranks are within each modality. The four classifiers of Chapter 3 are combined with five class-weighted additions (Extra-Trees, AdaBoost, ridge, SGD-logistic, linear SVM, fixed defaults). XGBoost leads on heart (rank 1) yet falls to rank 7 on lung; the linear family does the reverse. Spearman rank correlation heart-vs-lung: $\rho = 0.13$ ($p = 0.73$).

Model	Family	Heart MAcc	Rk	Lung ICBHI	Rk
XGBoost	boosting	**0.903**	**1**	0.500	7
SVM (RBF)	kernel	0.869	2	**0.537**	**1**
AdaBoost	boosting	0.868	3	0.503	6
Random forest	bagging	0.831	4	0.472	8
Ridge	linear	0.831	5	0.533	2
Linear SVM	linear	0.830	6	0.525	5
Logistic reg.	linear	0.825	7	0.533	3
SGD-logistic	linear	0.819	8	0.528	4
Extra-Trees	bagging	0.810	9	0.470	9

4.2 Deep cross-modal transfer and joint multi-task experiments

To probe whether the shared log-mel pipeline enables feature transfer between modalities, we ran three deep-learning experiments beyond the per-modality training of Chapter 3, each repeated over the same three seeds (1, 2, 42) and reported as mean±std: (i) direct transfer, where an encoder pre-trained on one modality is fine-tuned and evaluated on the other; (ii) joint multi-task training, where a single shared encoder with two task-specific heads is trained on both modalities at once; and (iii) in-domain references trained under the identical (non-HPO) protocol, so that the transfer comparison is strictly like-for-like. The transfer matrix in Figure 4.1 summarises the EfficientNet-B0 results; Table 4.2 gives all per-setting scores with their seed standard deviations.

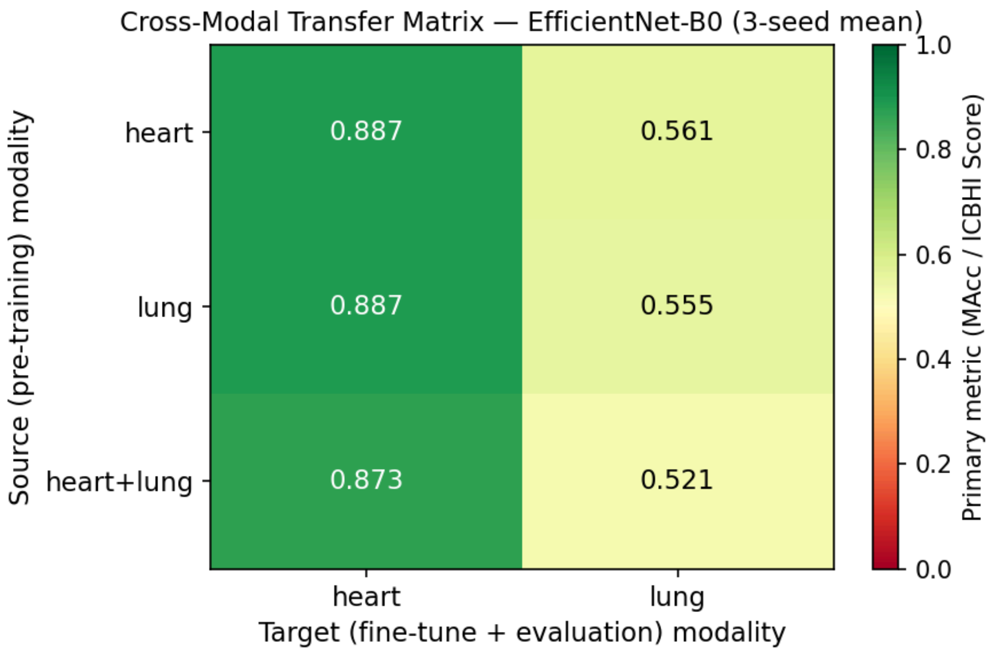

Figure 4.1 – Cross-modal transfer matrix for EfficientNet-B0 (three-seed means). Each cell shows the primary metric (MAcc for a heart target; ICBHI Score for a lung target) when a model whose features originate in the source modality (rows) is fine-tuned and evaluated on the target modality (columns): the diagonal is in-domain, the off-diagonal cells are direct transfer, and the bottom row is the joint multi-task model. SmallCNN values appear in Table 4.2.

The central result is that, once seed variance is taken into account, no cross-modal transfer penalty is *detectable* at three seeds in either direction; the experiment cannot rule out small effects below the seed spread, but it does rule out the strong asymmetry a single seed had suggested. Lung→heart transfer reaches MAcc = 0.887 ± 0.004 (EfficientNet-B0), indistinguishable from the in-domain heart reference (0.887 ± 0.010); heart→lung transfer reaches ICBHI = 0.561 ± 0.009, at or slightly above the in-domain lung reference (0.556 ± 0.003). The SmallCNN behaves identically (lung→heart 0.854 ± 0.004 vs. in-domain 0.852 ± 0.008; heart→lung 0.531 ± 0.013 vs. in-domain 0.526 ± 0.037). In short, with full fine-tuning on the target the choice of source modality is largely washed out: neither direction incurs a penalty detectable above seed noise, and the apparent 0.005 gains from opposite-modality pre-training are well within that noise and are not read as a benefit.

This corrects an artefact of single-seed evaluation. A single representative run had suggested a strong asymmetry, with heart→lung transfer apparently falling well below the lung in-domain baseline; that gap did not survive three-seed averaging and lay within the seed spread. The episode is a concrete illustration of why every deep result in this study is reported over multiple seeds: a lone run can manufacture a qualitative "finding" — here, transfer asymmetry — that dissolves under seed-robust evaluation. The robust cross-modal conclusion is therefore the classical one of the previous section: it is method *rankings*, not learned features, that fail to transfer between modalities.

The joint multi-task model (shared encoder, two heads) reaches MAcc $= 0.873 \pm 0.011/0.848 \pm 0.008$ for heart (EfficientNet-B0 / SmallCNN) and ICBHI $= 0.521 \pm 0.018/0.554 \pm 0.014$ for lung. Relative to per-modality training the joint model trades off slightly differently for the two backbones: it holds heart within about one standard deviation of in-domain and shows a small lung drop for EfficientNet-B0, whereas for the SmallCNN it slightly improves lung at a negligible heart cost. No catastrophic interference appears in either case, consistent with a shared encoder that has enough capacity to serve both tasks but cannot fully optimise for each at once.

Table 4.2 collects all per-setting scores for direct reference.

Table 4.2 – Per-modality deep-model scores under in-domain, transfer and joint settings, as mean±std over three seeds $(1, 2, 42)$. Transfer scores are placed in the columns of the *target* modality. In-domain references use the same non-HPO protocol as the transfer runs; the HPO-tuned in-domain values of Chapter 3 are slightly higher. The classical best is the per-modality recording/cycle-level score.

Setting	Heart CNN	Heart EffNet	Lung CNN	Lung EffNet
In-domain	0.852±0.008	0.887±0.010	0.526±0.037	0.556±0.003
Transfer heart→lung	—	—	0.531±0.013	0.561±0.009
Transfer lung→heart	0.854±0.004	0.887±0.004	—	—
Joint multi-task	0.848±0.008	0.873±0.011	0.554±0.014	0.521±0.018
Per-modality classical best	0.903	—	0.537	—

4.3 Audio Spectrogram Transformer: a transformer baseline

As a fourth deep model, a pretrained Audio Spectrogram Transformer (AST) [17], originally trained on AudioSet, was fine-tuned on both modalities under the same pipeline and, like the other deep models, evaluated as a mean ± standard deviation over three seeds $(1, 2, 42)$. One qualification still frames the result: the AST is heavyweight (86 million parameters, against four million for EfficientNet-B0) and, under the project's compute budget, each seed was fine-tuned for only a few epochs (two on heart, eight to ten on lung), so its scores are an epoch-limited figure rather than the model's ceiling.

On heart sounds the AST reaches MAcc $= 0.864 \pm 0.006$ at a recall-leaning operating point (Se $=$ 0.928, Sp $=$ 0.800): it favours recovering abnormal recordings at the cost of more false positives, a screening-friendly bias produced by class-weighted training. That this reflects genuine discrimination rather than over-fitting is confirmed by a high ROC-AUC (0.950 averaged over the three seeds). Its balanced MAcc nonetheless trails the tuned classical XGBoost (0.903) and the three-seed EfficientNet-

B0 (0.898), so the transformer does not displace the simpler heart models at this data scale and tuning budget (Figure 4.2).

On lung sounds the picture differs: the AST reaches ICBHI = 0.594 ± 0.023, the single best lung score in the study, ahead of the three-seed EfficientNet-B0 mean (0.555 ± 0.016) and every classical model. The lead is consistent rather than a lucky seed: even the weakest AST seed (0.576) exceeds the EfficientNet-B0 mean. This agrees with the literature position (Chapter 1) that transformer backbones with large-scale audio pretraining lead on ICBHI, and it indicates the audio-pretrained AST has a real, if modest, edge on the harder four-class respiratory task.

The AST thus both confirms that transformer audio models integrate into the shared pipeline without modification and, on lung sounds, edges ahead of the convolutional models over three seeds; training each seed to convergence at a larger epoch budget is the natural next step.

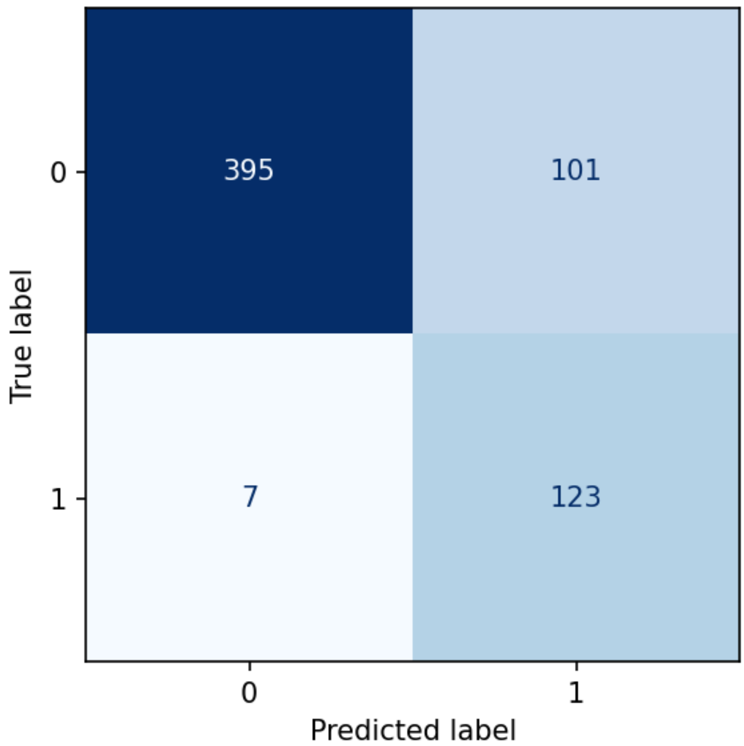

Figure 4.2 – Confusion matrix of the fine-tuned AST on heart sounds for a representative seed (seed 42: MAcc = 0.871, Se = 0.946), at the recording level over 626 test recordings. The recall-leaning operating point shows as near-complete recovery of the abnormal class at the cost of more false positives on the normal class.

4.4 Arterial sounds: an analytical sub-study

Arterial bruits occupy the third corner of this project's scope, and their absence from the empirical experiments requires careful documentation.

4.4.1 Acoustic characteristics of carotid bruits

A carotid bruit is produced when blood accelerates through a region of arterial stenosis, generating turbulent flow whose kinetic energy is partly radiated as acoustic pressure. The acoustic basis of this phenomenon was established by the foundational phonoangiography work of Lees and Dewey [26], who showed that the spectrum of a bruit recorded at the skin surface matches that of laboratory turbulent pipe flow and can be related quantitatively to arterial diameter and flow velocity. The sound is audible with a stethoscope placed at the lateral neck during systole, the half-cycle in which cardiac ejection drives peak flow velocity. In spectral terms, bruits occupy roughly 50–800 Hz, with the dominant energy below 400 Hz for mild stenoses and extending upward as stenosis severity increases and peak flow velocity rises. Duncan et al. [27] demonstrated that this upward spectral shift is systematic enough to estimate the residual lumen diameter directly from the bruit spectrum, a quantitative, feature-based reading of an auscultation signal that closely anticipates the classification approach pursued in this project for heart and lung sounds. This overlap with the heart-sound band (20–400 Hz) means that bruit detection from a stethoscope placed at the precordium (as opposed to the neck) is problematic without careful device placement and directional filtering.

The clinical motivation for automating this assessment is strong. Manual auscultation for carotid bruits has only modest, operator-dependent diagnostic accuracy: the Rational Clinical Examination review by Sauvé et al. [28] found the presence of a bruit to be at best a weak-to-moderate predictor of high-grade stenosis, with sensitivity and specificity that vary widely between examiners. Yet an audible bruit carries genuine prognostic weight: a meta-analysis pooling 17 295 patients reported that carotid bruits roughly double the rate of cardiovascular death and myocardial infarction [29]. An objective, reproducible acoustic classifier could therefore both reduce inter-observer variability and surface a clinically actionable risk marker, which is precisely the contribution that the feature-engineering and deep-learning pipelines compared in this report are designed to make for the heart and lung modalities.

The clinical pipeline for bruit assessment shares structural elements with the heart- and lung-sound pipelines studied here: the signal is captured at fixed sampling rates (typically 4000–10 000 Hz with modern digital stethoscopes), bandpass-filtered to the bruit-relevant band, segmented in cardiac-cycle- synchronised windows, and feature-extracted for classification. A recent smartphone study [4] demonstrated on-device deep-learning detection of carotid stenosis from neck auscultation, establishing proof-of-concept for machine-learning-based screening. An earlier computer-assisted auscultation system [5] was built specifically to acquire carotid vascular sounds reproducibly (a prerequisite for assembling a database of pathological bruits) yet, like the smartphone work, it released no open dataset.

4.4.2 Why supervised learning is not yet feasible

The obstacle to empirical work in this study is entirely a data-availability problem, not an algorithmic one. Every published carotid bruit system relies on proprietary recordings that were not released alongside the paper [4,5]. At the time of writing, an exhaustive search of PhysioNet, Zenodo, Figshare and GitHub yielded no openly licensed, cycle-labelled dataset of arterial bruit recordings suitable for training a supervised classifier. This absence is documented explicitly so that readers can verify the claim: if the situation changes, the pipeline of Chapter 2 would extend to arterial bruits with only two changes: the bandpass cutoffs (adjusted to 50–800 Hz or a tighter passband as appropriate) and the class label scheme (normal vs. bruit, or graded by stenosis severity). All other pipeline stages (resampling, windowing, MFCC / log-mel feature extraction, grouped (subject-level) splitting, evaluation) apply without modification.

4.4.3 What a future arterial dataset would require

For the unified pipeline to apply to a third modality, a suitable dataset would need to meet the same standards as CinC 2016 and ICBHI 2017: (i) at least several hundred recordings from independent subjects; (ii) ground-truth labels verified by duplex ultrasound (the reference standard for carotid stenosis); (iii) open licensing permitting unrestricted research use; and (iv) sufficient class balance, or at least a patient-independent split with documented label distribution. The methodological infrastructure to evaluate such a dataset is already in place in the shared pipeline described in Chapter 2.

4.4.4 A synthetic proof-of-concept

To show that this infrastructure does run end-to-end on arterial-band audio, and not merely in principle, we built a small *synthetic* dataset and passed it through the unchanged pipeline. No clinical data are used or claimed. The generator produces two classes of artificial neck recordings at 4000 Hz: a “bruit” class, in which a systolic-gated band-limited component models the turbulent-flow spectrum of the Lees-Dewey phonoangiography model [26], and a “normal” class carrying a comparable systolic component. The two classes are deliberately made hard to separate: both always contain mid-band energy of overlapping strength, and the only systematic difference is a subtle upward shift of the centre frequency in the bruit class (as a tighter stenosis raises the peak flow velocity and the spectral peak), with overlapping per-class distributions and a strong added noise floor. The signal is then processed by exactly the pipeline stages of Chapter 2, only the passband is changed to the arterial 50–800 Hz band: Butterworth bandpass, peak normalisation, fixed 3-second windows and the feature-set-B descriptor, with a strict patient-level split (84 train / 36 test synthetic subjects, 480 recordings) and the same zero-leakage assertion.

Under this deliberately overlapping contrast the classical models reach a recording-level MAcc of about 0.80 (logistic regression 0.799, random forest 0.783, RBF-SVM 0.740; Figure 4.3), comfortably above the chance level of 0.5 but far from the trivial separability of a cleanly separable toy task. The result carries no clinical weight whatever: the data are synthetic and the difficulty is set by hand. Its only claim is operational, that the unified pipeline ingests, splits, features and classifies arterial-band audio without modification beyond a passband change, exactly as the analytical sub-study argues it would for a future real dataset.

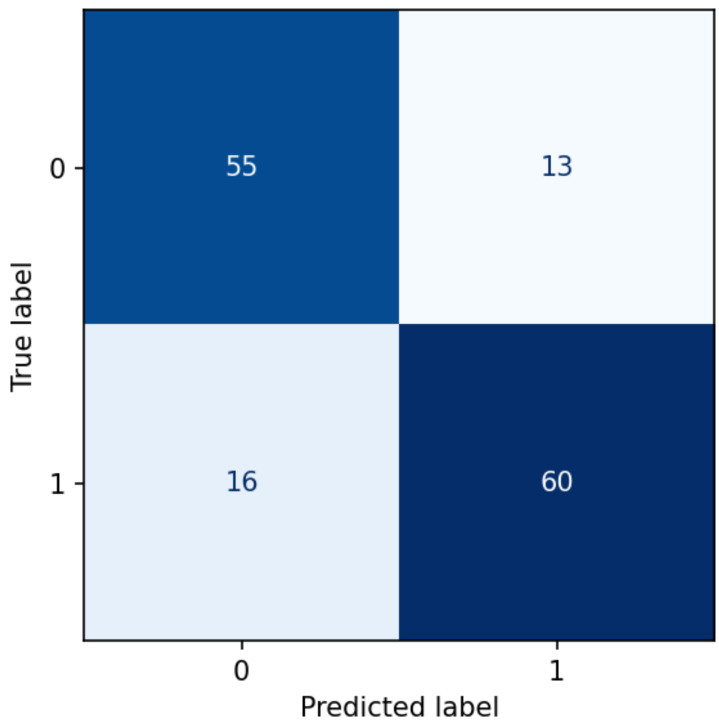

Figure 4.3 – Confusion matrix of the best classical model (logistic regression) on the *synthetic* arterial proof-of-concept (recording level, 144 test recordings, patient-level split). The data are artificial and the difficulty is set by hand; the figure illustrates only that the pipeline runs end-to-end on arterial-band audio, not any clinical accuracy.

4.4.5 Chapter summary

The cross-modal analysis delivers three findings. First, classical method rankings do not fully transfer: XGBoost dominates on heart sounds but falls to third place on lung (and to rank 7 of 9 once the panel is widened), while SVM is uniformly competitive. The Spearman rank correlation is weak and non-significant whether measured on the four core classifiers ($\rho = 0.60$ / 0.00 / 0.24 for set A / B / pooled) or on an expanded nine-classifier panel ($\rho = 0.42$ / 0.13 / 0.32; all $p > 0.19$), and the expansion reveals a model-family-by-modality interaction (tree ensembles favour heart, linear models favour lung) that underlies the non-transfer. A paired McNemar test confirms the inversion at the level of individual predictions. Second, deep cross-modal transfer, evaluated over three seeds, is benign and roughly symmetric: under full fine-tuning, both lung→heart and heart→lung transfer reach in-domain performance, so learned features carry over either way. A single-seed run had suggested an asymmetry, but it did not survive multi-seed averaging, a cautionary result that reinforces the study's emphasis on seed-robust evaluation. Joint multi-task training holds both modalities near per-modality performance with only small, backbone-dependent trade-offs and no catastrophic interference. Third, with HPO-tuned deep models, the deep-vs-classical gap on heart closes to within noise, while lung remains equally hard for all method families. These results motivate per-modality model selection even within a shared pipeline and quantify the cost of a naive one-model-all-modalities deployment.

The fine-tuned AST confirms that transformer audio models integrate cleanly into the pipeline and, on lung sounds, gives the single best score in the study over three seeds (ICBHI $= 0.594 \pm 0.023$); training each seed to convergence at a larger epoch budget is left to future work.

The arterial sub-study establishes only that the pipeline runs end-to-end on arterial-band audio — an engineering check, not evidence of generalisation — demonstrated on a synthetic benchmark (MAcc ≈ 0.80 on a hand-tuned contrast with no clinical weight), while empirical clinical work remains blocked by data unavailability, a gap that the community should address through a coordinated open data release.

Conclusion

This project set out to compare, honestly and reproducibly, classical and deep-learning methods for diagnosing cardiorespiratory pathologies from auscultation sound, under a single leakage-safe protocol spanning two modalities. The objectives stated in the Introduction were met as follows.

A reproducible, configuration-driven pipeline shared by the heart and lung modalities was assembled, with fixed random seeds, pinned dependencies, leakage-safe grouped splits (patient-level for lung, recording-level for heart) and an explicit zero-leakage assertion executed before every training run. On heart sounds the best classical configuration, XGBoost with the MFCC+Δ+ΔΔ + spectral-statistics feature set, reached MAcc = 0.903 (Se = 0.946, Sp = 0.859). After HPO tuning (128-trial bounded random search, val-only selection), EfficientNet-B0 reaches MAcc = 0.898 ± 0.008 (three seeds), comparable with the best classical model: the gap of 0.005 falls within one standard deviation, so deep learning narrows the classical advantage to within seed noise on heart sounds once the tuning effort is equalised, though XGBoost remains numerically best. A paired McNemar test on recording-level predictions makes the point concrete: the two are statistically indistinguishable at the seed where EfficientNet-B0 reaches 0.898 ($p = 0.68$), while XGBoost is significantly ahead on the weaker seeds, so neither family is reliably best. On lung sounds the models were evaluated at the cycle level with the official ICBHI score. The best configuration, HPO-tuned EfficientNet-B0, achieves ICBHI = 0.555 ± 0.016, marginally ahead of the best classical model (SVM-B, 0.537) and within the same performance tier. The four-class imbalanced structure limits all methods approximately equally on this task.

The cross-modal analysis, the project's principal novelty, combined ranking comparisons, direct transfer experiments and a joint multi-task probe. Classical method rankings do not transfer reliably: XGBoost dominates on heart but falls to third on lung, while SVM is competitive on both modalities (the heart–lung rank correlation is weak and non-significant whether measured on the four core classifiers (Spearman $\rho = 0.60$ / 0.00 / 0.24 for set A / B / pooled) or on an expanded nine-classifier panel ($\rho =$ 0.42 / 0.13 / 0.32, all $p > 0.19$), where XGBoost falls from rank 1 on heart to rank 7 of 9 on lung). The widened panel also exposes a model-family-by-modality interaction, tree ensembles favouring heart and linear models favouring lung, that underlies the non-transfer. A paired McNemar test on the individual predictions confirms the inversion is real rather than a rounding effect: XGBoost is significantly ahead of SVM on heart ($\chi^2 = 17.5$, $p \approx 1.5 \times 10^{-5}$), yet on lung its raw-accuracy edge does not convert into the balanced-metric win, which SVM holds. Deep cross-modal transfer, by contrast, shows no penalty detectable above seed noise once evaluated over three seeds: under fine-tuning both directions reach in-domain performance, and a transfer asymmetry suggested by a single seed did not survive multi-seed averaging, a cautionary result that underscores the value of seed-robust evaluation. Joint multi-task training holds both modalities near per-modality performance with only small, backbone-dependent trade-offs. It is thus method *rankings*, not learned features, that fail to transfer; modality-specific classifier selection remains necessary even within a shared pipeline.

An extension to the CirCor DigiScope 2022 dataset, on a genuinely patient-level split, added a finer heart task, murmur detection, and reversed the classical-versus-deep ordering seen on CinC: the deep SmallCNN reached MAcc = 0.810 ± 0.007, clearly ahead of the best classical model (0.688), showing that which method family wins is itself task-dependent. The reversal is carried by the SmallCNN specifically (it outscores the larger EfficientNet-B0, which had no CirCor-specific tuning), so the claim is that *a* deep model, not deep learning categorically, wins the finer task. The arterial sub-study established that the pipeline runs end-to-end on a third modality — an engineering check, not evidence of generalisation — demonstrated on a synthetic arterial-band benchmark (MAcc ≈ 0.80 on a hand-tuned, purely synthetic contrast with no clinical weight), while empirical clinical work is blocked by the absence of any openly licensed, labelled dataset of arterial bruits. A pretrained Audio Spectrogram Transformer (AST) was fine-tuned on both modalities over three seeds: it gave the single best lung score in the study (ICBHI = 0.594 ± 0.023) and a recall-leaning heart model (MAcc = 0.864 ± 0.006, Se = 0.928); on heart it does not displace the EfficientNet-B0 and classical baselines.

The work's positive outcomes are a leakage-safe, fully reproducible comparison across two modalities and an original cross-modal analysis; its negative outcomes are recorded honestly below.

Limitations

The following limitations bound every claim in this report.

- *Dataset scope.* Only two open datasets were used (PhysioNet/CinC 2016 and ICBHI 2017). Results may not transfer to recordings from other devices, populations or acquisition conditions.
- *No clinical validation.* The study is a methodological comparison on retrospective public data. No claim is made about clinical accuracy, safety, diagnostic equivalence to a physician, or readiness for deployment. The intended framing throughout is screening and triage support, not diagnosis.
- *Label noise and class imbalance.* Both datasets are imbalanced and carry annotation noise; the smallest lung class ("both") is especially scarce, which limits how reliably it can be learned and reported.
- *No open arterial data.* Arterial-bruit diagnosis could only be treated analytically, because no suitable open dataset exists; the corresponding clinical claims are conceptual, not empirical. The synthetic proof-of-concept demonstrates only that the pipeline runs end-to-end on arterial-band audio; its data are artificial, its difficulty is set by hand, and it carries no clinical weight.
- *Scope of deep models.* The deep-learning comparison covers a compact CNN, an EfficientNet-B0 transfer model and a fine-tuned Audio Spectrogram Transformer. The AST was trained over three seeds but for only a few epochs per seed under the compute budget, so its scores (best on lung, recall-leaning on heart) are a lower bound on its ceiling, limited by the epoch budget rather than by seed variance.

- *Binary heart tasks.* Both heart tasks are binary: normal versus abnormal on CinC and murmur present versus absent on CirCor. Neither grades or localises a specific pathology (for example individual murmur timing, shape or grade), which the richer CirCor metadata would in principle support but which is left to future work.
- *Split granularity for heart.* The public CinC 2016 release provides no complete recording-to-subject mapping, so the heart split is grouped at the recording level rather than the patient level. This eliminates window-level leakage (all windows of a recording stay on one side) but cannot rule out the same subject contributing recordings to both sides. The ICBHI lung split and the CirCor heart-murmur split are both genuinely patient-independent, so the patient-level CirCor result complements the recording-level CinC evaluation.
- *Single-split evaluation.* Every headline metric is computed on one held-out split per dataset (a single seeded grouped split for heart, the one official split for lung); the deep ±std values vary the seed on that fixed split, not the split itself. No outer cross-validation was run, so the absolute numbers carry split-selection variance that is not quantified, and the classical point estimates have no resampled confidence interval beyond the binomial interval reported for the best heart model.
- *No human baseline.* The screening framing implies a comparison against a non-specialist listener, but no human-expert benchmark was collected. The report therefore makes no claim of parity with, or superiority to, a clinician; the comparisons are strictly between automated methods.

Future work

Several extensions follow naturally. The Audio Spectrogram Transformer, here fine-tuned for only a few epochs per seed, should be trained to convergence at a larger epoch budget to confirm whether its three-seed lead on lung sounds and its recall-leaning heart behaviour widen against the EfficientNet-B0 baseline. Richer heart-sound labels (for example the murmur metadata of the CirCor DigiScope database) would turn the binary task into graded murmur classification. The joint multi-task architecture studied here could be scaled with larger shared encoders to close the remaining heart trade-off. Finally, if an openly licensed arterial-bruit dataset becomes available, the existing pipeline could be applied to a third modality with only a change of passband and configuration, completing the cardiorespiratory-and-arterial scope of the project's title.

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Annex A Full metric tables

This annex reproduces the complete per-model metric tables for both modalities, beyond the rounded summary in Chapter 3: the heart-sound classical results are given in Table A.1 and the lung-sound classical results in Table A.3, with the corresponding deep-learning models in Table A.2 and Table A.4. Numbers are taken directly from `results/tables/metrics_heart_classical.csv`, `metrics_lung_classical.csv`, `metrics_heart_cnn.csv` and `metrics_lung_cnn.csv`.

A.1 Heart-sound metrics (classical models)

Table A.1 – Complete heart-sound classical metrics. MAcc = (Se + Sp)/2. AUC-ROC is recording-level after majority vote. Feature sets: A = MFCC+Δ+ΔΔ (240-d); B = MFCC+Δ+ΔΔ+spectral stats (250-d).

Set	Model	MAcc	Se	Sp	macro-F1	AUC
A	LogReg	0.7945	0.8692	0.7198	0.7062	0.8494
A	SVM	0.8589	0.9154	0.8024	0.7827	0.9330
A	RF	0.8169	0.6923	0.9415	0.8270	0.9418
A	XGBoost	0.8791	0.9154	0.8427	0.8158	0.9500
B	LogReg	0.8248	0.9077	0.7419	0.7339	0.8985
B	SVM	0.8694	0.9385	0.8004	0.7882	0.9352
B	RF	0.8313	0.7231	0.9395	0.8370	0.9450
B	XGBoost	**0.9025**	**0.9462**	**0.8589**	0.8394	**0.9573**

A.2 Heart-sound metrics (deep-learning models, HPO-tuned, 3-seed mean)

Table A.2 – Heart-sound deep-learning metrics. HPO-tuned (val-only selection, 128 trials); values are the mean over three seeds (1, 2, 42). Parameter counts reflect the HPO-selected architecture (the tuned SmallCNN uses wider 32–256 channels).

Model	Params	MAcc	Se	Sp	macro-F1	AUC
SmallCNN	389 314	0.871 ± 0.009	0.910	0.831	0.805	0.946
EfficientNet-B0	4 010 110	0.898 ± 0.008	0.936	0.860	0.837	0.966

A.3 Lung-sound metrics (classical models)

Table A.3 – Complete lung-sound classical metrics. ICBHI Score = ($Se_{abnormal}$ + Sp)/2. Se_{crk}, Se_{wh}, Se_{both} = per-class sensitivities on crackle, wheeze and both classes.

Set	Model	ICBHI	Se	Sp	Se_{crk}	Se_{wh}	Se_{both}
A	LogReg	0.5069	0.6561	0.3577	0.357	0.340	0.140
A	SVM	0.5363	0.6180	0.4545	0.404	0.405	0.105
A	RF	0.4763	0.1013	0.8513	0.055	0.094	0.000
A	XGBoost	0.5108	0.4703	0.5513	0.292	0.244	0.140
B	LogReg	0.5329	0.6933	0.3724	0.397	0.349	0.163
B	SVM	**0.5368**	0.6152	0.4583	0.405	0.394	0.116
B	RF	0.4722	0.0976	0.8468	0.066	0.102	0.023
B	XGBoost	0.5002	0.3690	0.6314	0.214	0.231	0.116

A.4 Lung-sound metrics (deep-learning models, HPO-tuned, 3-seed mean)

Table A.4 – Lung-sound deep-learning metrics. HPO-tuned (val-only selection, 128 trials); values are the mean over three seeds (1, 2, 42). Per-class sensitivities vary across seeds; representative-run breakdowns appear in Chapter 3.

Model	Params	ICBHI	Se	Sp	macro-F1
SmallCNN	98 148	0.540 ± 0.022	0.755	0.325	0.301
EfficientNet-B0	4 012 672	**0.555 ± 0.016**	0.509	0.601	0.309

Annex B Volumetric characteristics and reproducibility

Annex 5 §2.5 requires volumetric characteristics. Table B.1 provides sample sizes, model sizes, training times and dataset volumes, derived from `results/tables/volumetrics_classical.csv` and `results/tables/volumetrics_cnn.csv`.

Table B.1 – Dataset and experiment volumetrics. Heart patient counts are omitted: CinC 2016 provides no recording-to-subject map, so the heart split is grouped at the recording level (Section 2.1).

Item	Heart	Lung
Dataset	CinC 2016 (A–E)	ICBHI 2017
Train recordings / patients	2 500 / —	551 / 79
Test recordings / patients	626 / —	369 / 47
Train segments/cycles	33 246	4 262
Test segments/cycles	4 167	2 636
Segment/cycle length	3.0 s (windows)	3.0 s (padded cycles)
Common sampling rate	4 000 Hz	4 000 Hz
Bandpass	20–400 Hz	200–1800 Hz
Filter order	Butterworth 4	Butterworth 4
Feature set A dimension	240-d	240-d
Feature set B dimension	250-d	250-d
Log-mel image size	64 × 128	64 × 128
Classical feature data (MB)	74.2	13.7
Log-mel data (MB)	1 226.8	226.2
SVM-B train time (s)	659	31
XGBoost-B train time (s)	21	26
SmallCNN parameters	389 314	98 148
SmallCNN train time (s)	262	26
EfficientNet-B0 parameters	4 010 110	4 012 672
EfficientNet-B0 train time (s)	1 345	204
Random seed	42	42

Pinned software environment. Python 3.11; librosa 0.11.0; scikit-learn 1.8.0; XGBoost 3.2.0; PyTorch 2.11.0; torchaudio 2.11.0; timm ≥ 1.0; imbalanced-learn 0.14.1; numpy ≥ 1.26; scipy ≥ 1.13; pandas ≥ 2.2. The full pinned dependency list is committed to the repository as `requirements.txt`.

Annex C Spectrogram and exploratory-analysis gallery

The following figures were generated by the exploratory data analysis stage of the pipeline (scripts that read the raw datasets and produce class-distribution, duration-histogram and example-spectrogram plots).

Heart-sound class distribution (Figure C.1): across the 3126 recordings of databases A–E, normal recordings dominate (2495, 79.8%) over abnormal ones (631, 20.2%); the A–E label files carry only the normal/abnormal verdict, so no recording falls in the organisers' residual "unsure" tier. This roughly four-to-one class imbalance motivates class-weighted training.

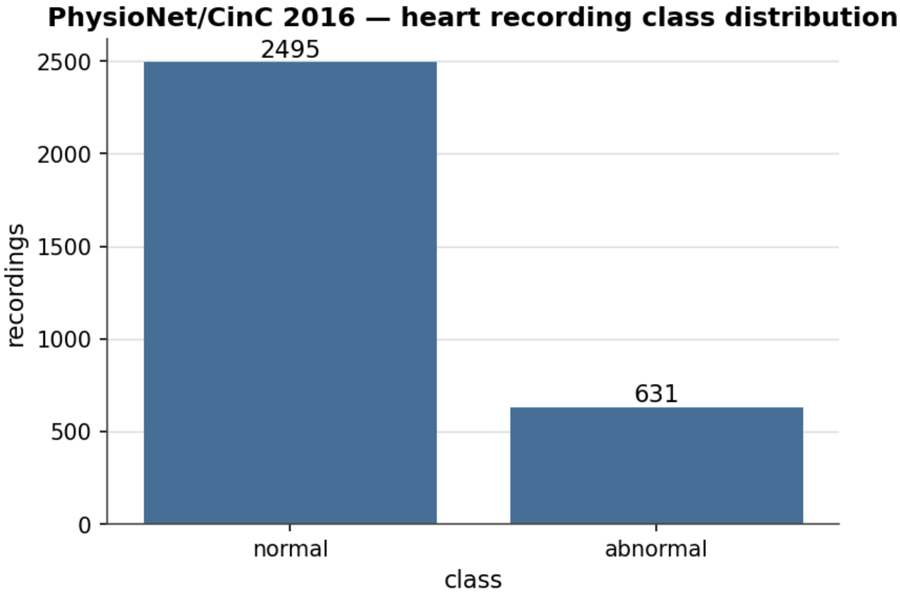

Figure C.1 – Class distribution in the CinC 2016 heart-sound training set (databases A–E combined): 2495 normal versus 631 abnormal recordings.

Heart recording duration (Figure C.2): recording lengths range from approximately 5 to 120 seconds, with a mode around 20–30 seconds. The 3-second windowing strategy in Chapter 2 generates between 1 and 40 windows per recording.

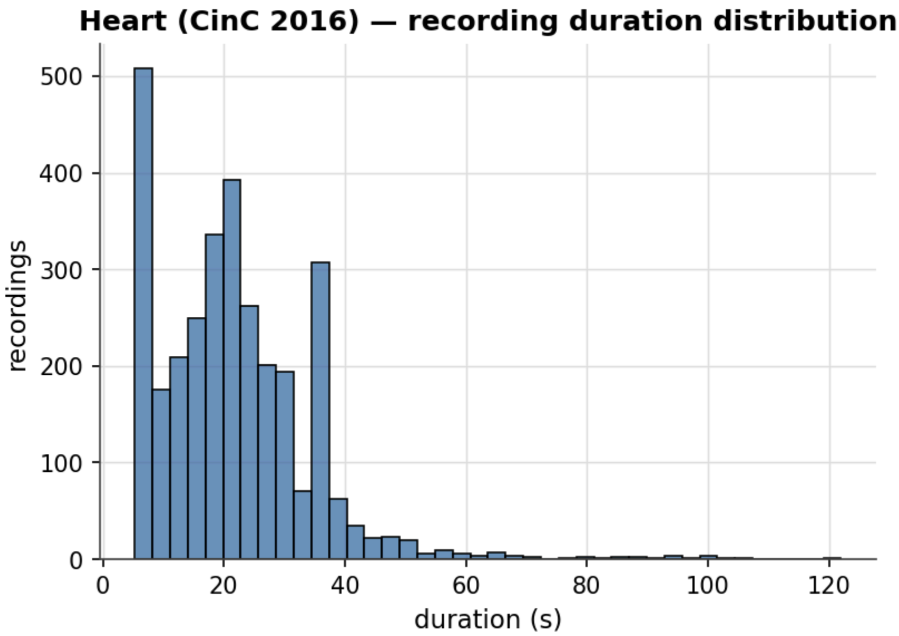

Figure C.2 – Duration distribution of CinC 2016 heart recordings.

Heart recordings per database (Figure C.3): database E dominates the pool with 2141 of the 3126 recordings (about 68%), followed by B (490) and A (409), while C (31) and D (55) contribute only a handful each. This heavy skew toward a single source database is a caveat for the heart results, and the bandpass filter must operate across this heterogeneous source mix.

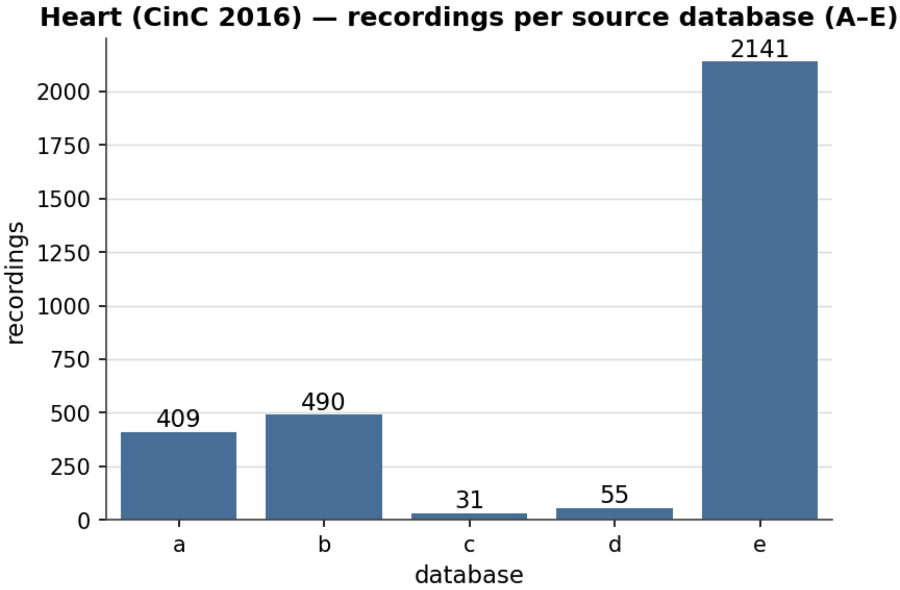

Figure C.3 – Recording counts per CinC 2016 source database (A–E).

Lung-sound class distribution (Figure C.4): the ICBHI 2017 cycle-level distribution shows that "normal" (52.8%) greatly outnumbers the three abnormal classes, with "both" (crackle + wheeze) being the rarest at 7.3%.

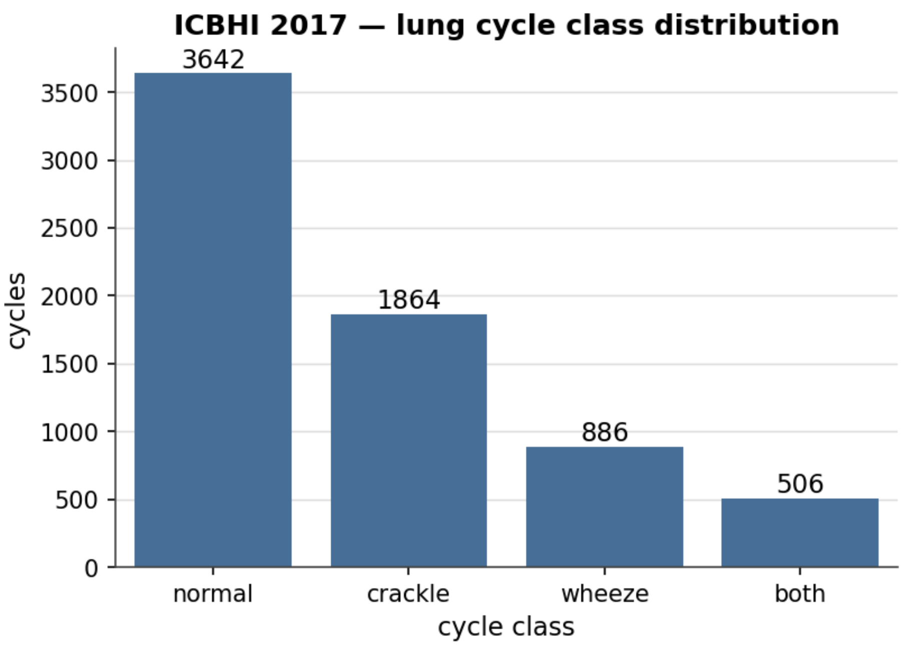

Figure C.4 – Cycle-level class distribution in ICBHI 2017 (6 898 annotated cycles).

Lung cycle counts per class (Figure C.5): the absolute per-class cycle counts underlying the distribution above, confirming the "both" class as the scarce minority that bounds four-class performance.

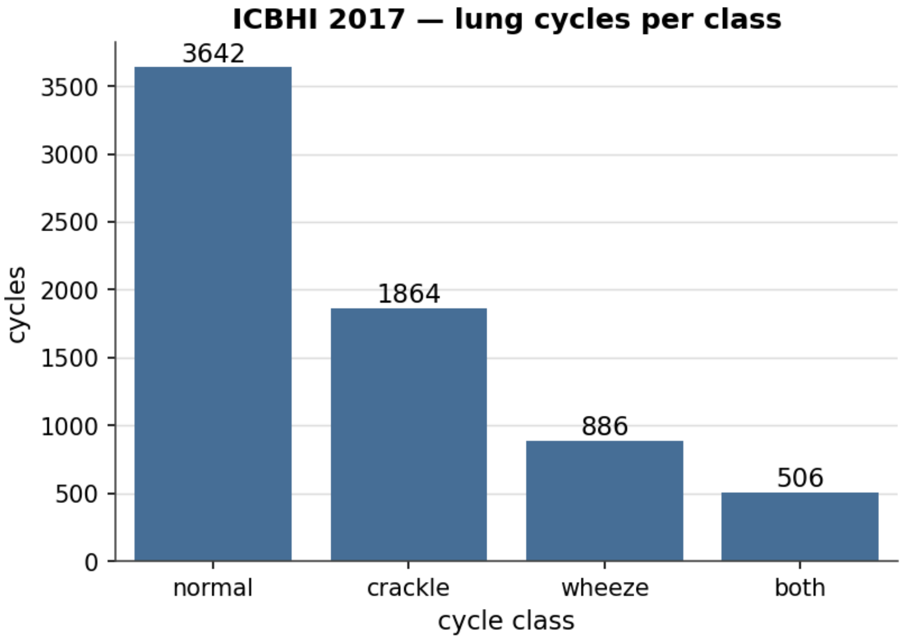

Figure C.5 – Absolute respiratory-cycle counts per ICBHI 2017 class (normal, crackle, wheeze, both).

Lung cycle duration (Figure C.6): annotated respiratory cycles vary in length; all are zero-padded or trimmed to a fixed 3.0-second window before feature extraction.

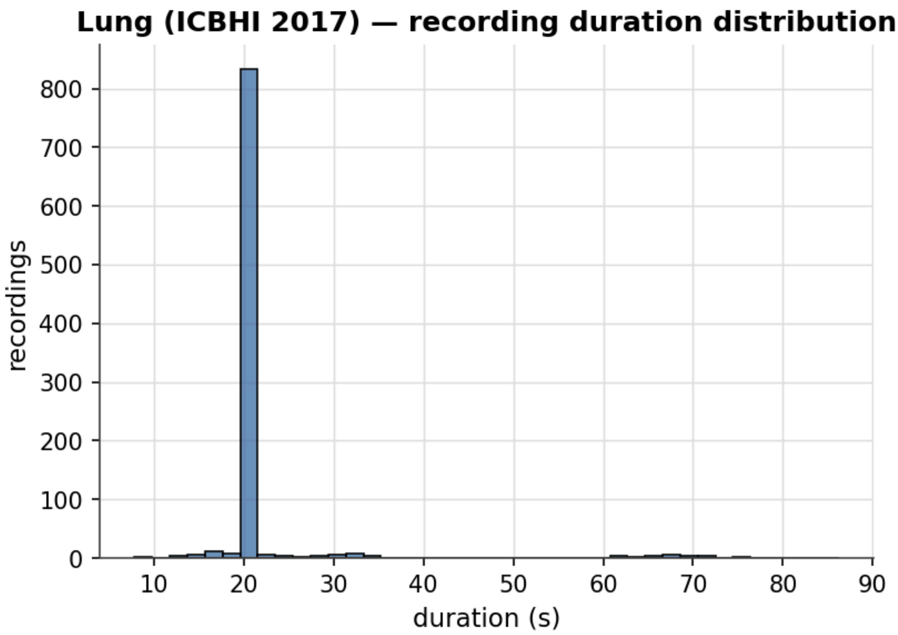

Figure C.6 – Distribution of annotated respiratory-cycle durations in ICBHI 2017. Cycles are padded or trimmed to 3.0 s before feature extraction.

Native sampling-rate histogram (Figure C.7): the four ICBHI recording devices contribute at 4000 Hz, 10 000 Hz, 22 050 Hz and 44 100 Hz. Resampling to 4000 Hz in preprocessing eliminates device-specific spectral artefacts.

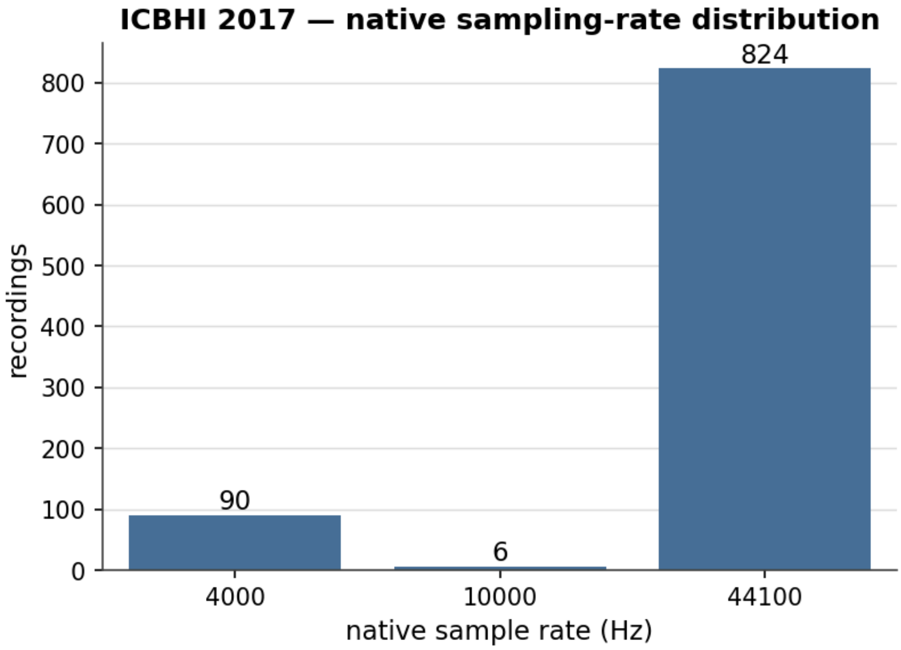

Figure C.7 – Native sampling-rate distribution across ICBHI 2017 recordings. All recordings are resampled to 4000 Hz before feature extraction.

Example spectrograms (heart, Figure C.8 and Figure C.9): the normal spectrogram shows clear S1–S2 periodicity concentrated below 150 Hz. The abnormal spectrogram shows a sustained mid-systolic murmur occupying a wider frequency band (the recording is labelled abnormal; CinC 2016 does not provide a specific murmur type).

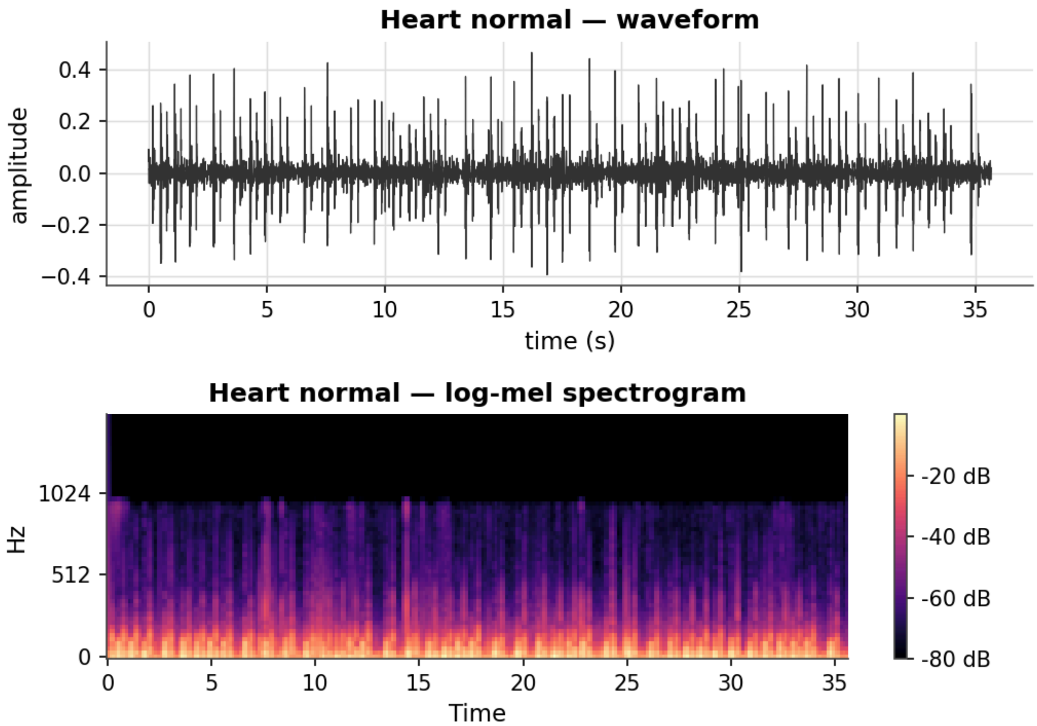

Figure C.8 – Example log-mel spectrogram panel: normal heart recording (CinC 2016). S1 and S2 events are visible as periodic vertical intensifications.

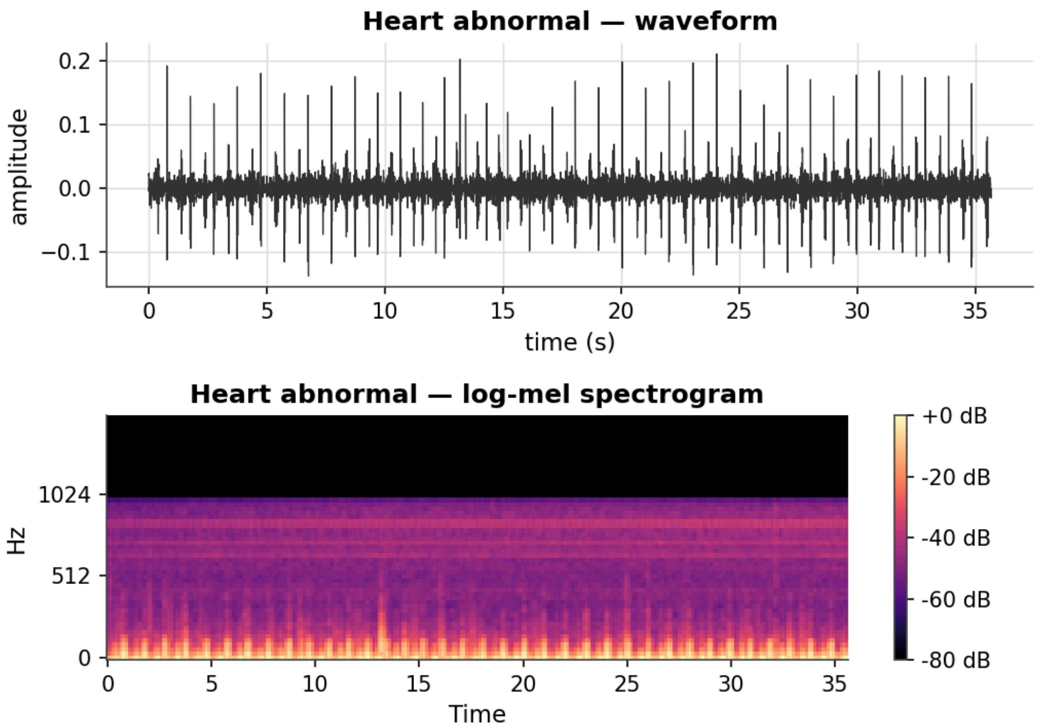

Figure C.9 – Example log-mel spectrogram panel: abnormal heart recording (CinC 2016) with a mid-systolic murmur. The murmur occupies a broad frequency band between the S1 and S2 events.

Example lung-cycle panels (Figure C.10): representative log-mel spectrograms for each of the four ICBHI 2017 cycle classes, illustrating the acoustic diversity the respiratory classifier must separate within a single short cycle.

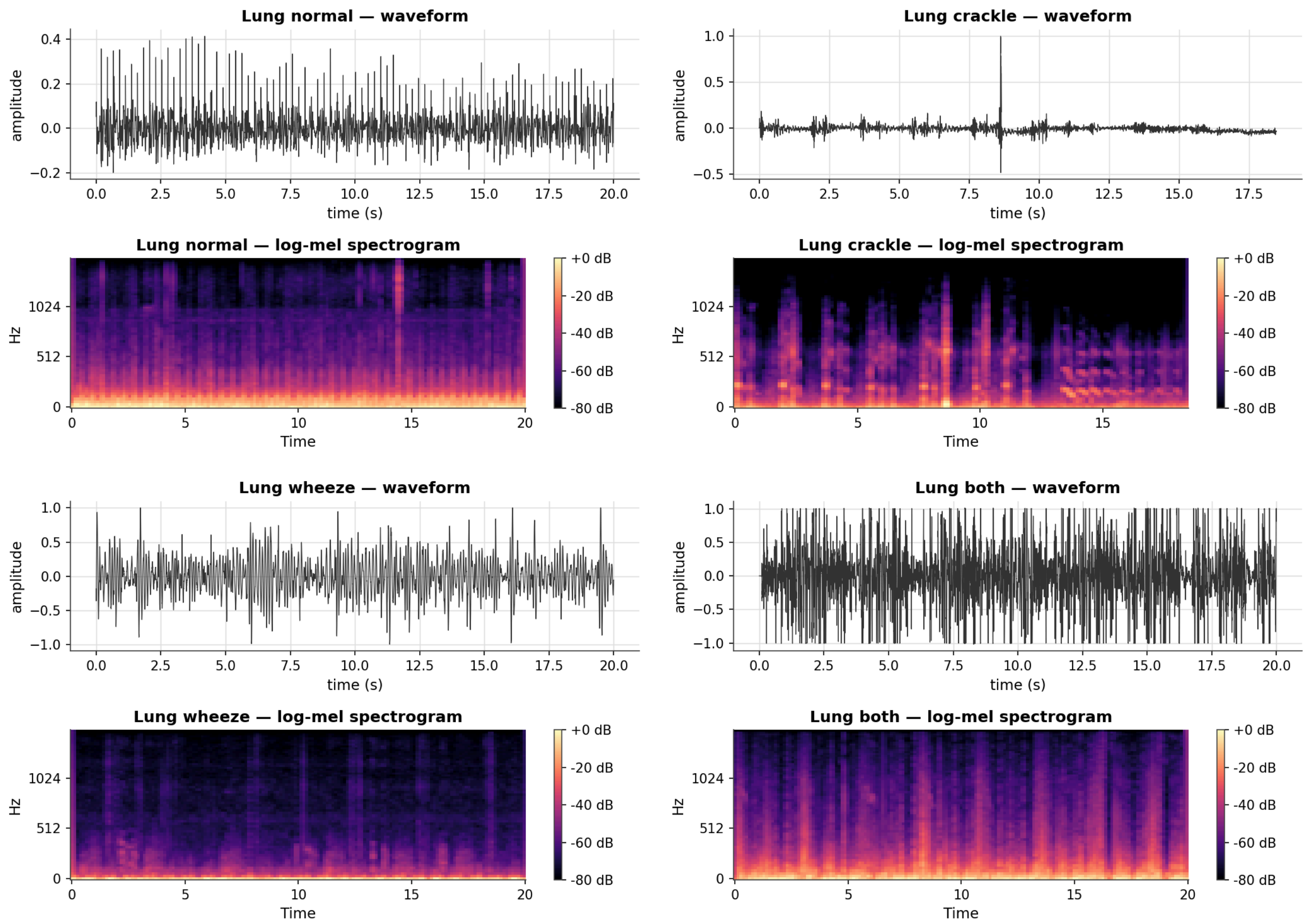

Figure C.10 – Example log-mel spectrogram panels for the four ICBHI 2017 cycle classes (top row: normal, crackle; bottom row: wheeze, and crackle-plus-wheeze “both”).

Annex D Supplementary confusion matrices and training curves

For completeness and transparency this annex collects the confusion matrices of every classical configuration — the main text (Chapter 3) shows only the best model per modality — together with the remaining deep-model training diagnostics. The two best classical models per modality (heart XGBoost-B and SVM-B; lung SVM-B) are repeated here so that each gallery is self-contained. Heart matrices are at the recording level (626 test recordings); lung matrices are at the cycle level over the four ICBHI classes (2636 test cycles). The heart classical matrices appear in Figure D.1 and Figure D.2, the lung classical matrices in Figure D.3 and Figure D.4, the remaining deep-model matrices in Figure D.5, and the remaining training curves in Figure D.6.

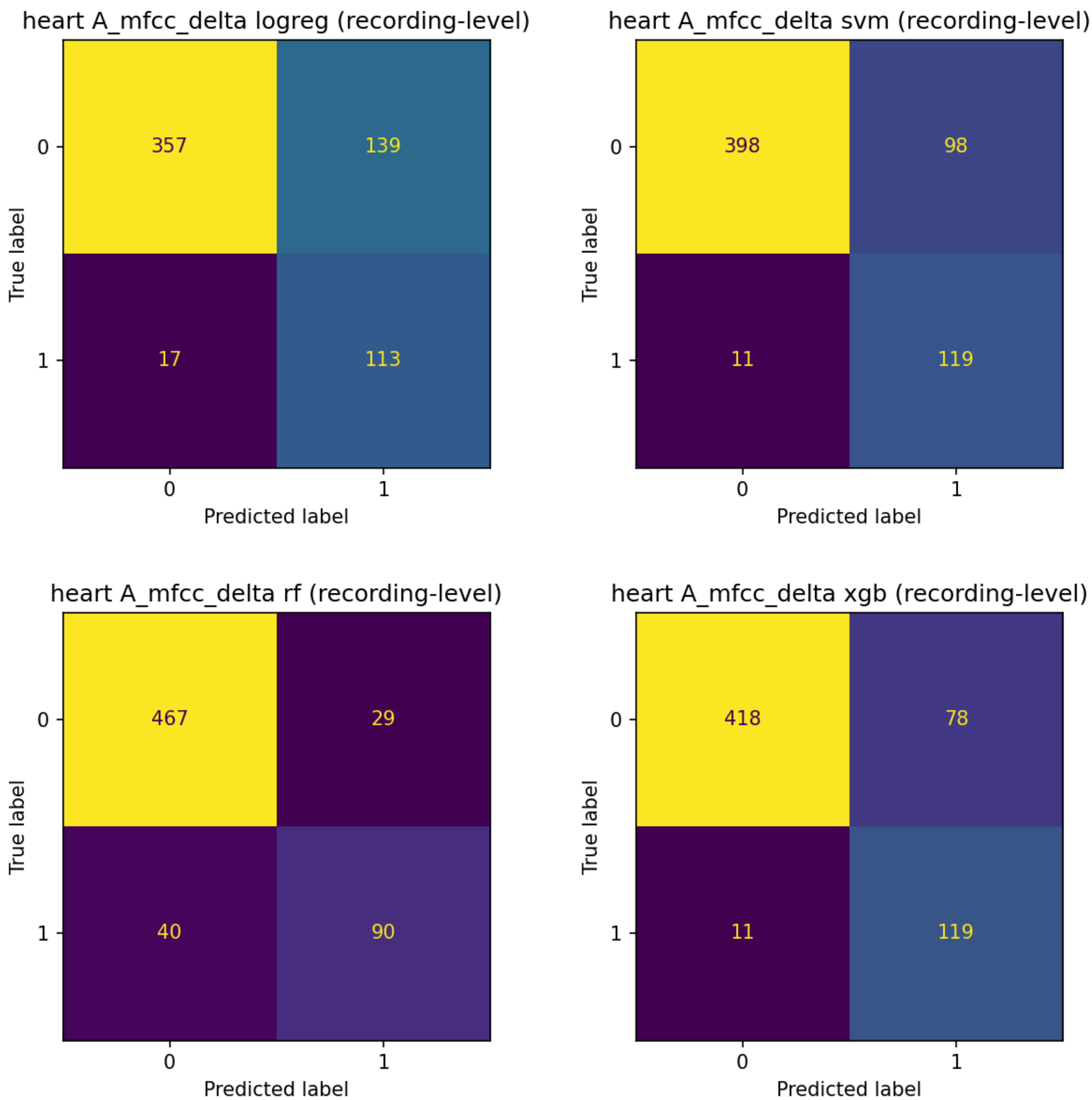

Figure D.1 – Heart-sound confusion matrices, feature set A (MFCC+Δ+ΔΔ). Top row: logistic regression, SVM; bottom row: random forest, XGBoost.

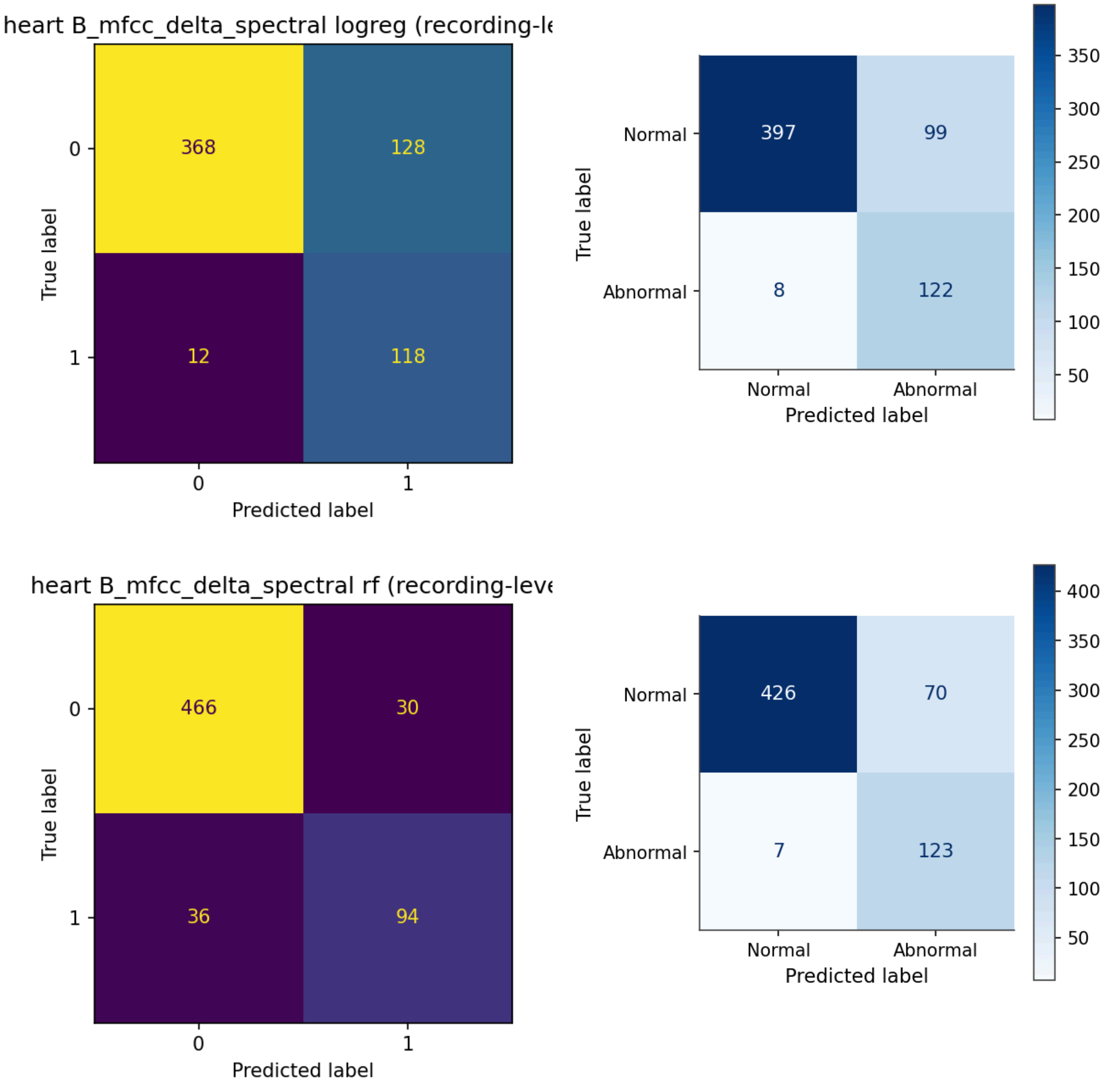

Figure D.2 – Heart-sound confusion matrices, feature set B (+ spectral statistics). Top row: logistic regression, SVM; bottom row: random forest, XGBoost (the best classical heart model, MAcc = 0.903).

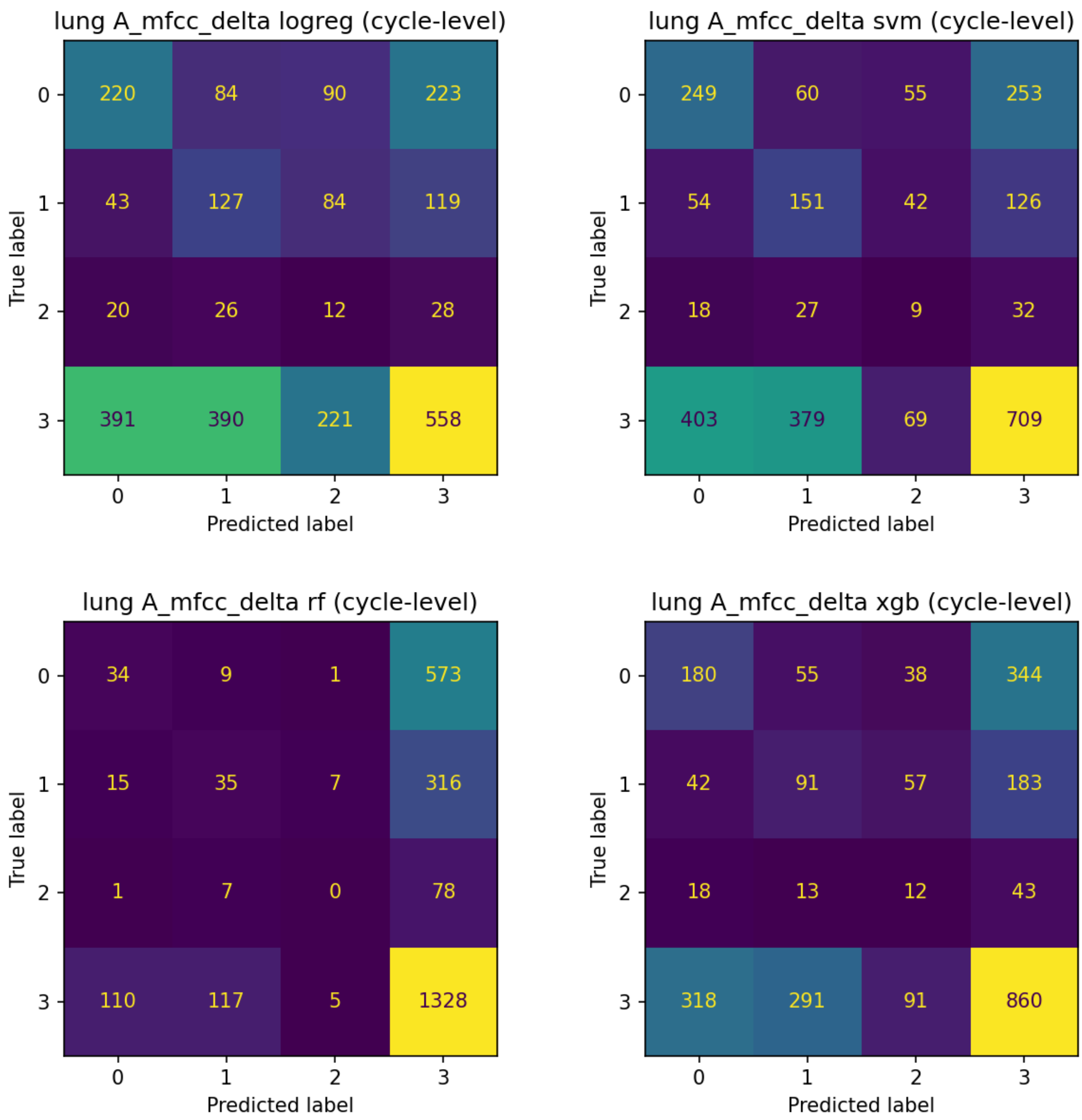

Figure D.3 – Lung-sound confusion matrices, feature set A (MFCC+Δ+ΔΔ). Top row: logistic regression, SVM; bottom row: random forest, XGBoost. The random-forest collapse toward the majority normal class is visible as a near-empty set of abnormal rows.

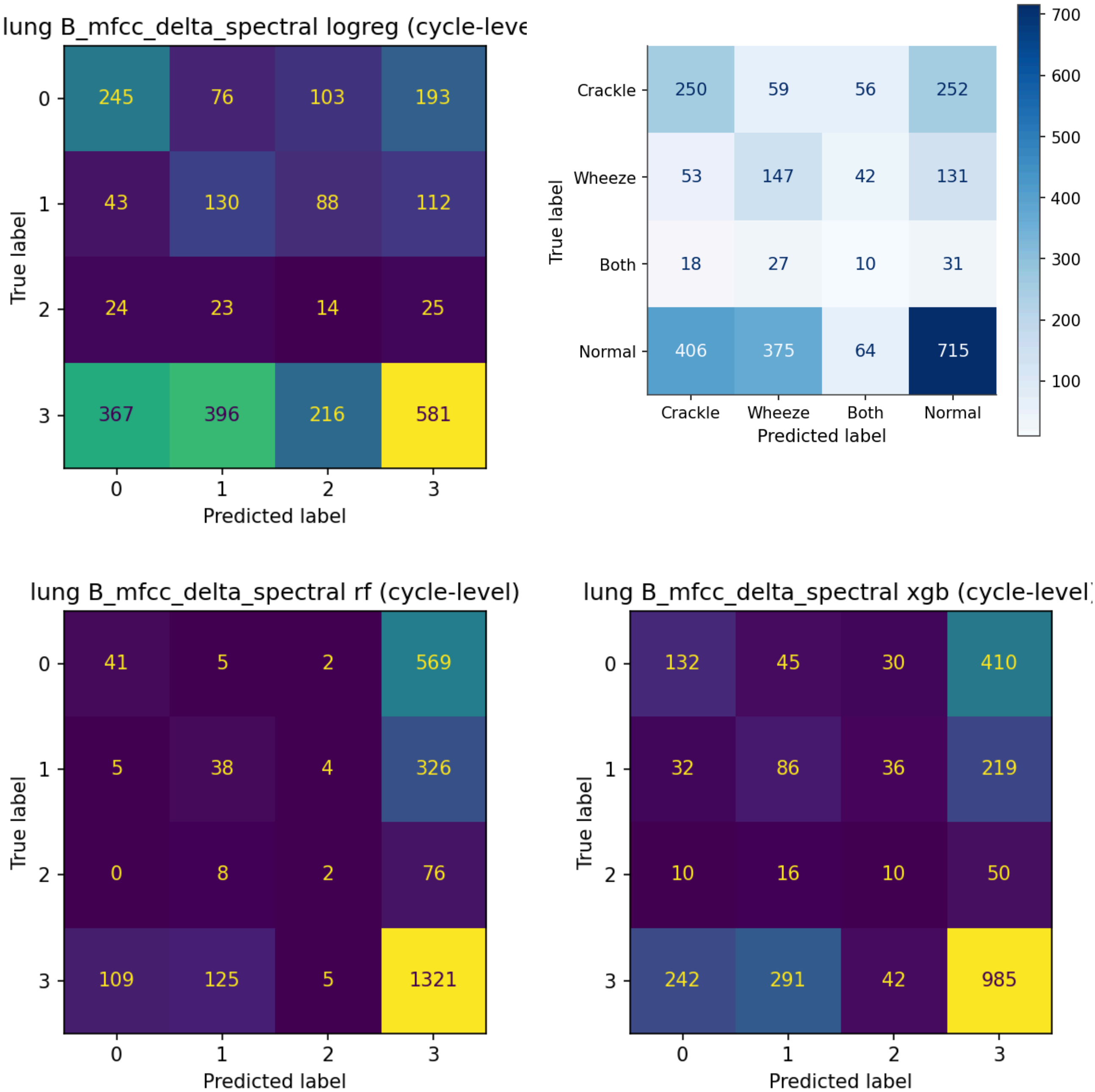

Figure D.4 – Lung-sound confusion matrices, feature set B (+ spectral statistics). Top row: logistic regression, SVM (the best classical lung model, ICBHI = 0.537); bottom row: random forest, XGBoost.

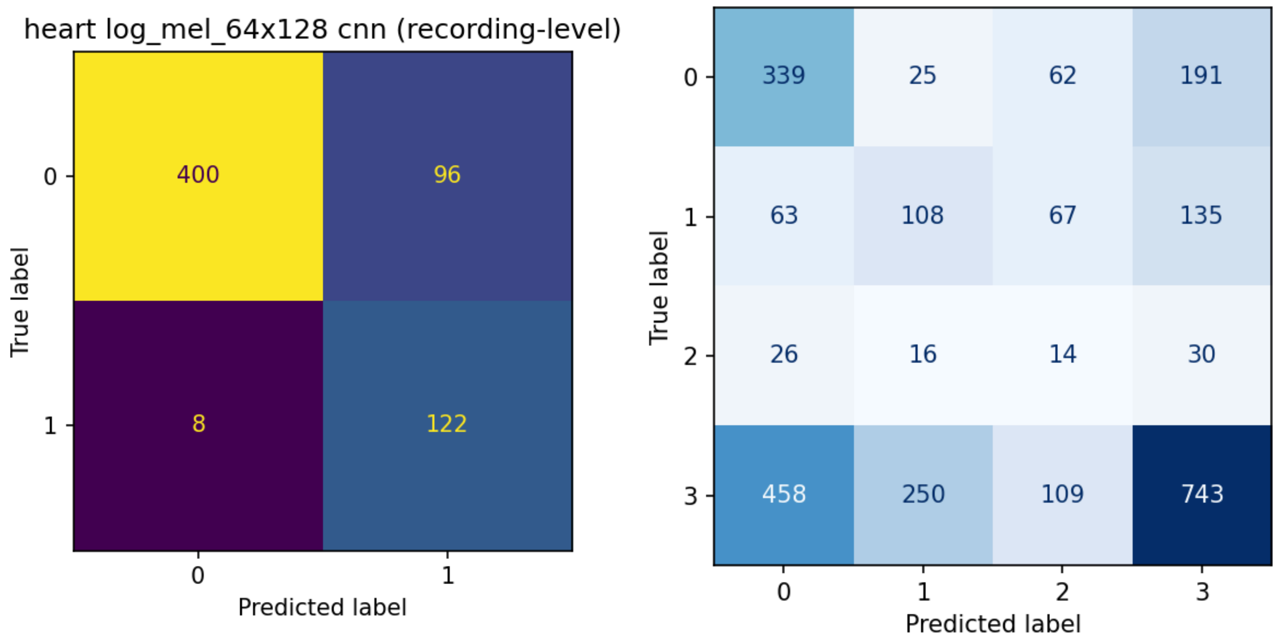

Figure D.5 – Remaining deep-model confusion matrices (single representative seed). Left: heart SmallCNN (recording level); right: lung Audio Spectrogram Transformer (cycle level). The best EfficientNet-B0 and the heart AST matrices appear in Chapters 3 and 4.

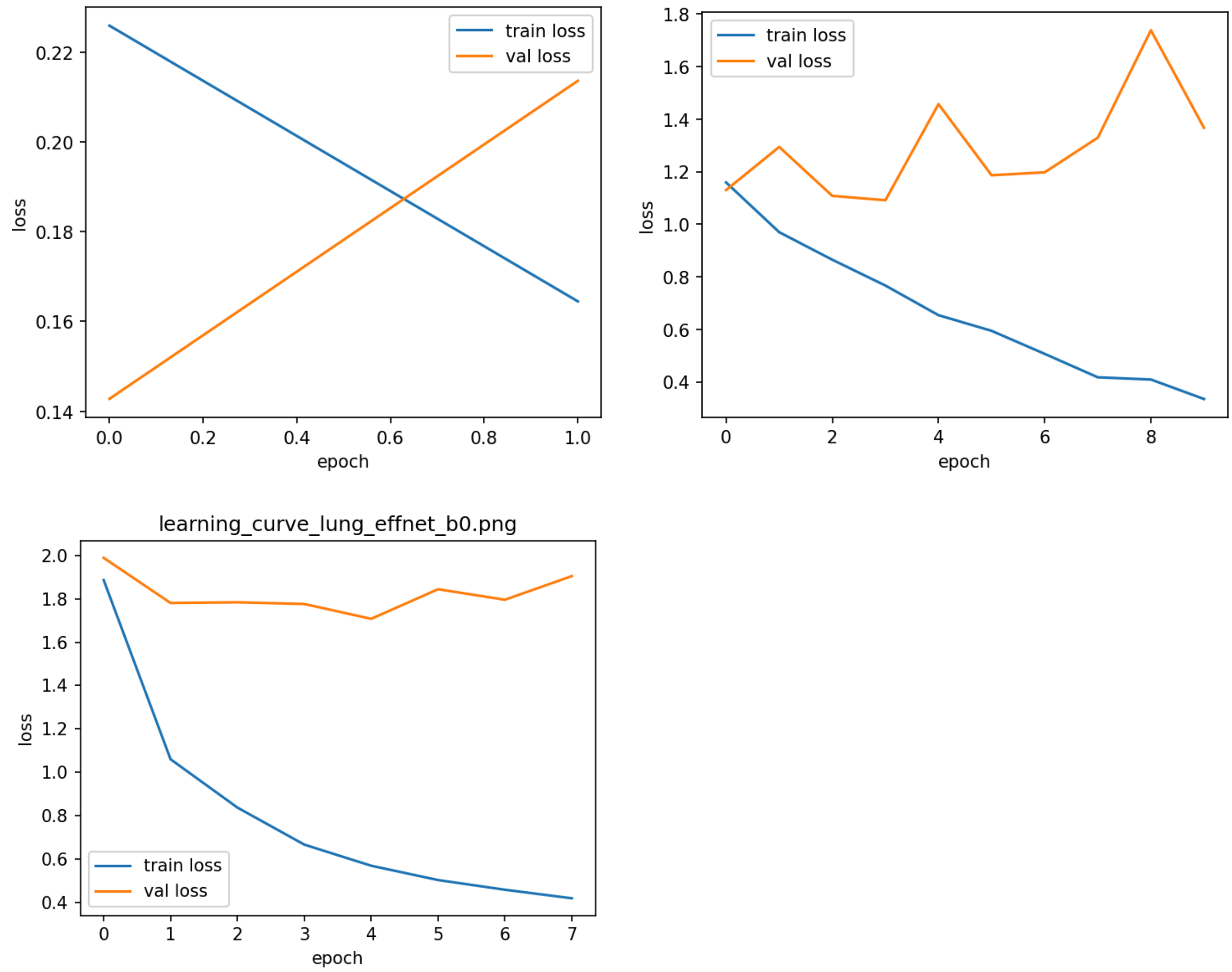

Figure D.6 – Remaining deep-model training curves (train versus validation primary metric): heart AST, lung AST and lung EfficientNet-B0. The SmallCNN and heart EfficientNet-B0 curves appear in Chapter 3; early stopping restores the best-validation checkpoint in every case.

Annex E Repository structure

The reproducible code accompanying this report is organised as follows (top-level, abridged):

```
params/{heart,lung}.yaml   per-modality preprocessing parameters
src/
  config.py                shared config; seeds RNGs at 42; canonical paths/
rates
  preprocess.py            resample → Butterworth bandpass → peak-normalise
  segment.py               fixed 3-s heart windows (silence/ragged-tail guards)
  features.py              MFCC+Δ+ΔΔ + spectral stats (240-d / 250-d vectors)
  spectrograms.py          torchaudio log-mel 64×128 image
  split.py                 patient-level splits + assert_no_patient_leakage
  metrics.py               MAcc, ICBHI score, majority vote, confusion matrices
  cnn.py                   SmallCNN + EfficientNet-B0 builders
  train_classical.py       classical fit→predict→evaluate driver
  train_cnn.py             deep-learning train+evaluate driver
```

```
  cross_modal.py            ranking transfer + joint multi-task model
scripts/                    CLI drivers (download → build → run_*)
tests/                      leakage / determinism / metric / pipeline tests
results/
  tables/unified_comparison.csv   one row per (modality, feature set, model)
  tables/metrics_{heart,lung}_{classical,cnn}.csv  full per-model metrics
  tables/volumetrics_{classical,cnn}.csv  dataset sizes, model params, times
  tables/mcnemar_classical.csv    paired McNemar tests (classical pairs)
  tables/mcnemar_vs_deep.csv      paired McNemar XGBoost vs EfficientNet (3
seeds)
  tables/effnet_heart_preds.csv   frozen per-seed EfficientNet predictions
(McNemar input)
  tables/ast_multiseed{,_raw}.csv AST mean±std and per-seed scores (3 seeds)
  tables/expanded_classifiers.csv nine-classifier panel + rank-transfer
Spearman
  tables/arterial_synthetic.csv   synthetic arterial proof-of-concept scores
  figures/                        confusion matrices, learning curves, EDA
report/
  main.typ                  master Typst document
  sections/                 section files (abstract, introduction, ch1–ch4,
                            conclusion, bibliography, annexes)
  refs.bib                  bibliography
  helpers.typ               shared formatting helpers
docs/pdf/                   compiled report PDF (the submission deliverable)
```

The repository link is supplied as a separate `TXT` file in the submission package, as required by the department.